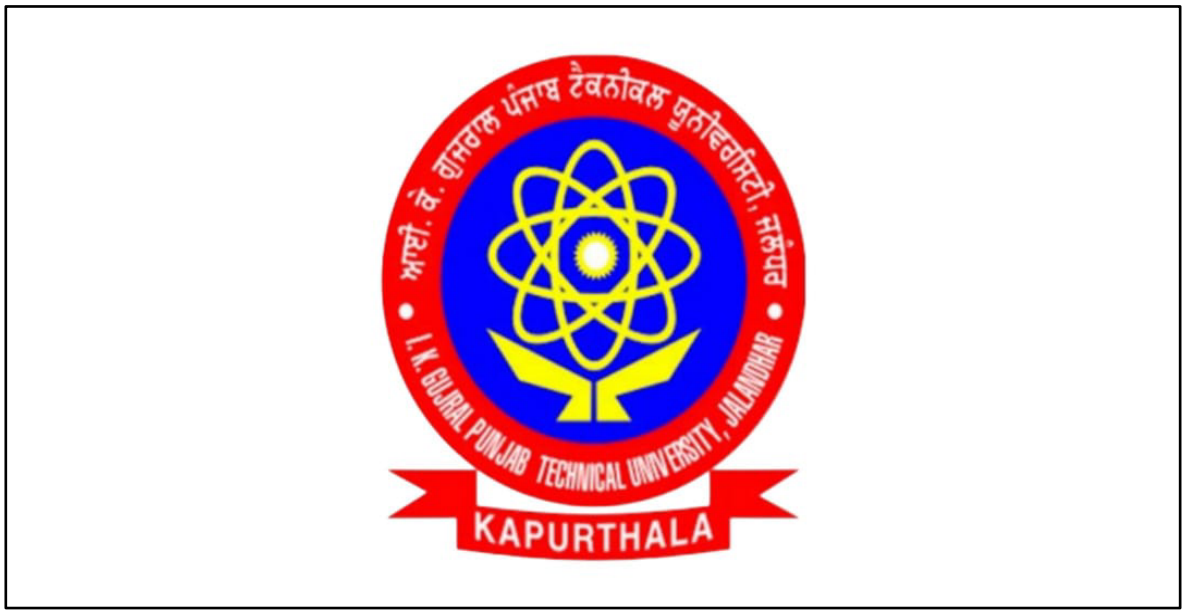
PROJECT REPORT

FLEET FUEL INTELLIGENCE : COST PREDICTION , EFFICIENCY SCORING AND ANOMALY DETECTION USING MACHINE LEARNING

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR MINOR PROJECT



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DECLARATION

I hereby declare that the project titled **"Fleet Fuel Intelligence: Cost Prediction, Efficiency Scoring and Anomaly Detection Using Machine Learning"** is a genuine and authentic record of the work carried out by me as part of the requirements for the Minor Project during the period from January 2026 to April 2026 for the award of the degree of Bachelor of Technology in Computer Science & Engineering at IK Gujral Punjab Technical University Campus, Hoshiarpur, under the guidance of Dr. Brijesh Bakariya.

I further declare that this work has not been submitted, either in part or in full, to any other institution or university for the award of any degree or diploma.

(Signature of Student)

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Date: 20 April 2026

Certified that the above statement made by the student is correct to the best of our knowledge and belief.

(Signature of Guide)

(Signature of Head of Department)
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ABSTRACT

This project presents an advanced machine learning-based system for fleet fuel intelligence including transaction analysis, cost prediction, efficiency scoring, and anomaly detection. The system is built using real-world fuel transaction data provided by Nurol Company and implements a multi-model pipeline integrated into an interactive web-based dashboard.

The proposed solution consists of three independent machine learning models. A cost prediction model is used to estimate the total fuel transaction amount based on features such as fuel quantity, mileage, and cost per kilometer. An efficiency scoring model evaluates the performance of each transaction on a scale of 0 to 100, helping identify efficient and inefficient fuel usage patterns. Additionally, an anomaly detection model is implemented as a classification system to identify suspicious or abnormal transactions using probability scores.

The system includes a complete data pipeline involving preprocessing, feature engineering, encoding techniques, and model training. Special features such as cost per kilometer and time-based attributes are derived to improve model performance. One-hot encoding with drop-first strategy is used for categorical variables.

A user-friendly dashboard is developed using Streamlit, allowing users to input transaction details and obtain real-time predictions. The system provides actionable insights through interactive outputs such as predicted cost, efficiency score, and anomaly detection results.

The results demonstrate that the system effectively improves fuel monitoring, enhances operational efficiency, and supports data-driven decision-making. This project highlights the practical application of machine learning in intelligent fleet management systems.

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Department of Computer Science & Engineering
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TABLE OF CONTENTS:-

Declaration	2
Abstract	3
Acknowledgement	4
Table of contents	5-8
Chapter 1: Introduction	
1.1 Background	9
1.2 Problem Statement	10
1.3 Objectives of the study	10-11
Chapter 2:Literature review	
2.1 Overview of machine learning in fuel analysis	12
2.2 Related work	12
2.3 Research gap	13
Chapter 3: System Overview	
3.1 Existing System	14
3.2 Limitations of existing system	14
3.3 Proposed System	14
3.4 Advantages of proposed system	15

TABLE OF CONTENTS:-

Chapter 4: Dataset Descrption	
4.1 Introduction to dataset	16
4.2 Source of dataset	16
4.3 Features description	16-17
4.4 Data characteristics	17
4.5 Data quality issues	17

Chapter 5: Tools and technologies used	
5.1 Introduction	18
5.2 Programming Language	18
5.3 Libraries and Frameworks	18
5.4 Development environment	18-19
5.5 Machine learning techniques	19
5.6 Data visualization tools	19
5.7 Hardware requirements	19-20
5.8 Software requirements	20
Chapter 6: Data Preprocessing	
6.1 Data Cleaning	21
6.2 Handling missing values	21
6.3 Data transformation	21
6.4 Data type conversion	21
Chapter 7: Feature Engineering	
7.1 Feature extraction	22

Chapter 5: Tools and technologies used	
7.2 Feature creation	22
7.3 Importance of features	22
Chapter 8: Exploratory data analysis (EDA)	
8.1 Distribution analysis	23-26
8.2 Relationship analysis	26-28
8.3 Correaltion analysis	28-29
8.4 Outlier detection	29-31
Chapter 9: Methodology	
9.1 Overview of methodology	32
9.2 Workflow of the system	32-33
9.3 Data preprocessing steps	34
Chapter 10: Machine Learning Models	
10.1 Introduction to machine learning	35
10.2 Cost prediction model (Regression)	35
10.3 Efficiency score model	35-36
10.4 Amomaly detection model (Isolation Forest)	36
Chapter 11: Model Evaluation	
11.1 Evaluation metrics	37
11.2 Performance analysis	37-38
Chapter 12: System Architecture	
12.1 Architecture diagram	39-40
12.2 Working flow	41

Chapter 5: Tools and technologies used	
Chapter 13: Data flow diagrams (DFD)	
13.1 DFD Level 0	42
13.2 DFD Level 1	42-43
13.3 DED Level2	43-44
Chapter 14:Entity relationship diagram(ER diagram)	
14.1 ER diagram description	45-47
Chapter 15: Results and Discussion	
15.1 Cost prediction results	48
15.2 Efficiency score results	49
15.3 Anomaly detection results	49-50
15.4 Interpretation of results	50-51
Chapter 16: Snapshots	
16.1 Screenshots	52-55
Chapter 17: Applications	
17.1 Industrial applications	56
17.2 Real world use cases	56-57
Chapter 18: Advantages and limitations	
18.1 Advantages	58
18.2 Limitations	58-59
Chapter 19: Future Scope	
19.1 Improvements	60
19.2 Future enhancements	61
Chapter 20:Conclusion	

CHAPTER 1: INTRODUCTION

1.1 Background:

Fuel management is a critical component in industries such as transportation, logistics, construction, and energy sectors, where large fleets of vehicles are involved. Efficient monitoring and control of fuel consumption are essential to reduce operational costs, improve performance, and prevent resource wastage.

In traditional fuel management systems, data is collected and stored but not effectively analyzed. These systems rely heavily on manual processes, making it difficult to identify patterns, detect anomalies, or generate meaningful insights. As a result, organizations often face challenges such as fuel inefficiency, unauthorized usage, and lack of transparency in operations.

With the rapid growth of data and advancements in technology, machine learning has emerged as a powerful tool for analyzing large datasets and extracting meaningful insights. Machine learning algorithms can automatically identify patterns, trends, and relationships within data, enabling organizations to make data-driven decisions.

In the context of fleet fuel intelligence, machine learning can be applied to:

- Analyze fuel transaction behavior
- Predict transaction costs based on operational features
- Evaluate fuel efficiency using scoring mechanisms
- Detect unusual or fraudulent transactions

This project utilizes real-world fuel transaction data provided by Nurol from operational fuel management systems to develop an intelligent and automated analysis platform. By applying machine learning techniques such as regression-based prediction, scoring models, and anomaly detection, the system aims to enhance fuel monitoring and improve operational efficiency.

The integration of machine learning and data analytics into fuel management systems represents a significant step toward automation and intelligent decision-making. This project demonstrates how modern data-driven approaches can be used to solve real-world problems in industrial fleet management systems.

1.2 Problem statement:

In modern fleet fuel management systems, large volumes of transaction data are generated daily. However, most traditional systems are limited to data storage and basic reporting, lacking advanced analytical capabilities. This results in inefficient utilization of fuel resources and difficulty in extracting meaningful insights from the data.
Organizations often face several challenges, including:
• Inability to predict fuel transaction costs accurately
• Lack of efficiency evaluation mechanisms
• Difficulty in detecting abnormal or fraudulent transactions
• Inefficient manual monitoring processes
Despite the availability of detailed transaction data, there is a lack of intelligent systems that can effectively analyze this data and support real -time decision-making.
Therefore, there is a need for an automated and intelligent system that can:
• Analyze large-scale fuel transaction data efficiently
• Predict transaction cost using machine learning
• Evaluate fuel efficiency through scoring techniques
• Detect anomalies and suspicious activities
This project addresses these challenges by developing a multi-model machine learning system integrated with a web-based dashboard for intelligent fuel analysis.

1.3 Objectives of the study:-

The main objective of this project is to develop an intelligent fleet fuel analysis system using machine learning techniques to improve operational efficiency and decision-making.
The specific objectives are as follows:

1. **To analyze fuel transaction patterns**
   Understand trends and behavior in fuel usage across different transactions.
2. **To predict transaction cost**
   Develop a regression model to estimate fuel cost based on input features.
3. **To evaluate efficiency using scoring model**
   Assign a score (0–100) to measure fuel usage efficiency.
4. **To detect anomalies in fuel transactions**
   Identify abnormal or suspicious transactions using classification

models.

5. **To perform feature engineering**
   Create derived features such as cost per kilometer and time-based attributes.
6. **To automate fuel data analysis**
   Replace manual processes with an intelligent automated system.
7. **To develop an interactive dashboard**
   Build a Streamlit-based application for real-time predictions.
8. **To support data-driven decision-making**
   Provide insights that help in optimizing fuel usage and reducing cost.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview of machine learning in fuel analysis

Machine Learning (ML) has become an important tool in analyzing large-scale industrial data, especially in sectors such as transportation, construction, and fleet management. In fuel analysis, ML techniques are used to process large volumes of transaction data and extract meaningful patterns that help improve efficiency and reduce operational costs.

Fuel consumption and transaction cost depend on multiple factors such as fuel quantity, distance, vehicle condition, and operational behavior. Traditional analytical methods are not capable of handling such complex data effectively. Machine learning algorithms can learn from historical data and identify hidden relationships.

In modern fleet systems, machine learning is used for:

- Cost prediction using regression models
- Efficiency evaluation using scoring systems
- Anomaly detection using classification models

These techniques improve automation, accuracy, and overall system performance.

2.2 Related Work

Several researchers have applied machine learning techniques in fuel analysis and transaction monitoring systems. Some studies have focused on regression models to predict fuel cost and consumption. These models provide accurate forecasting and help in planning operational resources.

Other research has used clustering techniques to group similar fuel usage patterns, helping in identifying efficient and inefficient operations. Anomaly detection methods such as Isolation Forest and classification algorithms have been widely used to detect irregular patterns and potential fraud.

Recent systems also integrate dashboards and visualization tools for real-time monitoring. However, many of these systems rely on single-model approaches and do not provide a comprehensive analysis framework.

2.3 Research gap

Despite advancements, several limitations still exist in current systems:
• Most systems use a single machine learning model instead of a multi-model approach
• Lack of efficiency scoring mechanisms
• Limited real-time interaction and visualization
• Poor integration of prediction, scoring, and anomaly detection
This project addresses these gaps by developing an integrated system that combines multiple machine learning models with an interactive dashboard for real-time analysis.

CHAPTER 3: SYSTEM OVERVIEW

3.1 Existing system

Existing fuel monitoring systems are mainly based on manual processes or basic analytical tools. These systems collect and store data but lack advanced capabilities for prediction, scoring, and anomaly detection.

The data is typically analyzed using spreadsheets or simple reports, which provide limited insights into fuel usage patterns. Such systems are not capable of handling large datasets efficiently and fail to provide real-time intelligence.

3.2 Limitations of existing system

• No cost prediction capability
• No efficiency scoring mechanism
• Weak anomaly detection
• High manual effort
• Limited scalability
• Lack of real-time insights

3.3 Proposed system

The proposed system is an intelligent fleet fuel analysis system based on a multi-model machine learning pipeline. The system consists of the following components:

• **Data Preprocessing:** Cleaning and preparing raw data
• **Feature Engineering:** Creating features such as cost per KM and time attributes
• **Machine Learning Models:**

- Cost Prediction Model

- Efficiency Scoring Model
- Anomaly Detection Model

• **Dashboard:**

Streamlit-based interface for real-time interaction

The system provides predictions, scores, and anomaly detection results in an integrated manner.

3.4 Advantages of proposed system

The proposed system offers several advantages over traditional systems:

- Automated analysis
- Improved prediction accuracy
- Real-time results
- Multi-model intelligence
- Better decision-making
- Scalable architecture

CHAPTER 4: DATASET DESCRIPTION

4.1 Introduction to dataset

The dataset used in this project is based on real-world fuel transaction and fleet-related data provided by Nurol Company from an operational fuel management system. It contains detailed information about fuel usage, transaction records, vehicle activity, mileage, and operational parameters. The dataset plays a crucial role in building and evaluating machine learning models for cost prediction, efficiency scoring, and anomaly detection.

The data is used to analyze fuel transaction patterns, evaluate efficiency, detect anomalies, and predict fuel transaction cost. Mileage is considered as an important parameter in the system, as it directly affects fuel efficiency and cost calculations. Since the dataset is derived from real-world fleet operations, it makes the project more practical and relevant for industrial applications. The dataset used in this project consists of a large number of fuel transaction records collected over a period of time. It contains approximately **10279 rows** and **20 columns**

The size of the dataset is sufficient to train machine learning models effectively and ensures reliable analysis. A larger dataset helps in improving model accuracy, capturing mileage-based patterns, and enhancing the overall performance of the system.

4.2 Source of dataset

The dataset used in this project was provided by **Nurol** from its fleet fuel monitoring and operational management system. The dataset contains real-time and recorded fuel transaction details collected from vehicles, fuel stations, and digital monitoring platforms.

The data includes information related to fuel quantity, mileage, transaction amount, station details, payment methods, and operational activities. Since the dataset is based on actual industrial operations, it improves the practical relevance and reliability of the proposed machine learning system.

4.3 Features description

The dataset consists of multiple features (columns) that describe different aspects of fuel transactions, vehicle activity, and operational behavior. Some of the important features include:

• Vehicle ID: Unique identifier for each vehicle
• Product Quantity: Amount of fuel consumed
• Total Amount: Total cost of fuel transaction
• Mileage: Distance covered per unit fuel (important for efficiency calculation)
• VAT Amount: Tax applied on fuel transactions
• Product Name: Type of fuel (e.g., petrol, diesel)
• Station Name: Fuel station where the transaction occurred
• Payment Method: Mode of payment used (cash, card, etc.)
• Transaction Time (Hour, Day, Month): Time-based attributes

These features are used as input variables for machine learning models to perform cost prediction, efficiency scoring, and anomaly detection. Mileage plays a key role in deriving features such as cost per kilometer and efficiency score.

4.4 Data characteristics

The dataset has the following characteristics:

• It is based on real-world operational data
• It includes both numerical and categorical features
• It contains time-based and mileage-based information
• It consists of a large volume of transaction data
• It may contain noise and inconsistencies due to real-world conditions

4.5 Data Quality issues

The dataset may contain certain data quality issues, such as:

• Missing values in mileage or transaction attributes
• Duplicate records

• Outliers in fuel consumption or mileage values
• Inconsistent or noisy data

These issues are handled during the data preprocessing stage to improve the accuracy and performance of machine learning models. Proper handling of mileage-related inconsistencies is important for accurate efficiency scoring and cost prediction.

CHAPTER 5: TOOLS AND TECHNOLOGIES USED

5.1 Introduction

This chapter describes the technologies, tools, and methodologies used in the development of the Fleet Fuel Intelligence System. The system is designed to analyze fuel transaction data, predict transaction cost, evaluate efficiency, and detect anomalies using machine learning techniques. Various software tools and libraries were utilized to ensure efficient data processing, model building, and visualization..

5.2 Programming Language

The primary programming language used in this project is Python. Python was chosen due to its simplicity, readability, and strong support for data science and machine learning. It provides a wide range of libraries that make data preprocessing, feature engineering, and model implementation efficient.

5.3 Libraries and Frameworks

Several Python libraries and frameworks were used in this project:

• Pandas: Used for data manipulation and preprocessing
• NumPy: Used for numerical computations
• Matplotlib & Seaborn: Used for data visualization
• Scikit-learn: Used for implementing machine learning models such as regression and classification
• Joblib: Used for saving and loading trained models

• Streamlit: Used to build an interactive web-based dashboard

These libraries help in building a complete machine learning pipeline from data preprocessing to deployment.

5.4 Development Environment

The development of this project was carried out using the following environment:• IDE: Visual Studio Code (VS Code)
• Operating System: Windows
• Python Version: Python 3.x

VS Code provides debugging support and extensions for efficient Python development.

5.5 Machine Learning Techniques

The project utilizes multiple machine learning techniques:

• Regression Model: Used for predicting fuel transaction cost
• Scoring Model: Used for evaluating efficiency based on mileage and fuel usage
• Classification Model: Used for anomaly detection and identifying unusual transactions

These techniques help in generating accurate predictions and meaningful insights.

5.6 Data Visualization Tools

Data visualization plays an important role in understanding patterns and insights. The following tools were used:

• Matplotlib: For basic plots
• Seaborn: For statistical visualizations
• Streamlit: For interactive dashboard visualization

These tools help in presenting outputs such as cost prediction, efficiency score, and anomaly results.

5.7 Hardware Requirements

The system requires the following minimum hardware specifications:

• Processor: Intel i3 or higher
• RAM: 4 GB or more
• Storage: At least 500 MB free space
• Internet connection (optional for deployment)

5.8 Software Requirements

The software requirements include:
• Operating System: Windows / Linux / macOS
• Python (Version 3.x)
• Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, streamlit, joblib
• IDE: Visual Studio Code or Jupyter Notebook

CHAPTER 6: DATA PREPROCESSING

6.1 Data cleaning

Data cleaning is the first step in preprocessing. The dataset may contain errors, duplicate records, or irrelevant data. These issues can affect model performance if not handled properly. In this step, unnecessary columns are removed, duplicates are deleted, and incorrect entries are corrected. This ensures that the dataset becomes consistent and suitable for analysis.

6.2 Handling missing values

Real-world datasets often contain missing values due to system errors or incomplete records.

To handle missing data:
• Numerical values (including mileage) are filled using mean or median
• Categorical values are filled using mode
• Records with excessive missing values may be removed

This improves model reliability and accuracy.

6.3 Data transformation

Data transformation converts raw data into a suitable format for machine learning models.

Techniques used include:
• Normalization for scaling values
• Standardization for uniform distribution

This ensures that features like fuel quantity and mileage contribute equally to the model.

6.4 Data type conversion

Different data types are converted into numerical form:

• Categorical variables (station, product, payment) are encoded
• Date-time values are converted into hour, day, and month
• Mileage and numeric values are properly formatted

This step makes the dataset compatible with machine learning

algorithms.

CHAPTER 7: FEATURE ENGINEERING

7.1 Feature extraction

Feature extraction involves selecting important features from the dataset that significantly affect predictions. Key features include fuel quantity, total amount, mileage, and transaction time. Irrelevant features are removed to improve model efficiency and reduce noise.

7.2 Feature creation

Feature creation is used to generate new features that improve model performance.

Some important derived features include:
• Cost per KM = Total Amount / Mileage
• Fuel Efficiency Score (based on mileage and fuel usage)
• Time-based features (hour, day, month)

These features help the models better understand patterns and improve prediction accuracy.

7.3 Importance of features

Feature creation is used to generate new features that improve model performance.

Some important derived features include:
• Cost per KM = Total Amount / Mileage
• Fuel Efficiency Score (based on mileage and fuel usage)
• Time-based features (hour, day, month)

These features help the models better understand patterns and improve prediction accuracy.

CHAPTER 8: EXPLORATORY DATA ANALYSIS (EDA)

8.1 Distribution analysis

Distribution analysis is performed to understand how the values of different numerical features are spread across the dataset. It helps in identifying the nature of data such as skewness, concentration, and variability.

In this project, distribution plots (histograms) are generated for key features such as **Product Quantity, Total Amount, and Mileage**. These graphs show how frequently different values occur in the dataset.

From the visual analysis, it is observed that:
• Product quantity is unevenly distributed, indicating variation in fuel consumption across transactions
• Total transaction amount shows a right-skewed distribution, meaning higher-value transactions are less frequent
• Mileage values vary significantly, reflecting differences in vehicle efficiency

These distribution graphs help in understanding the overall behavior of the dataset and are useful for further modeling.

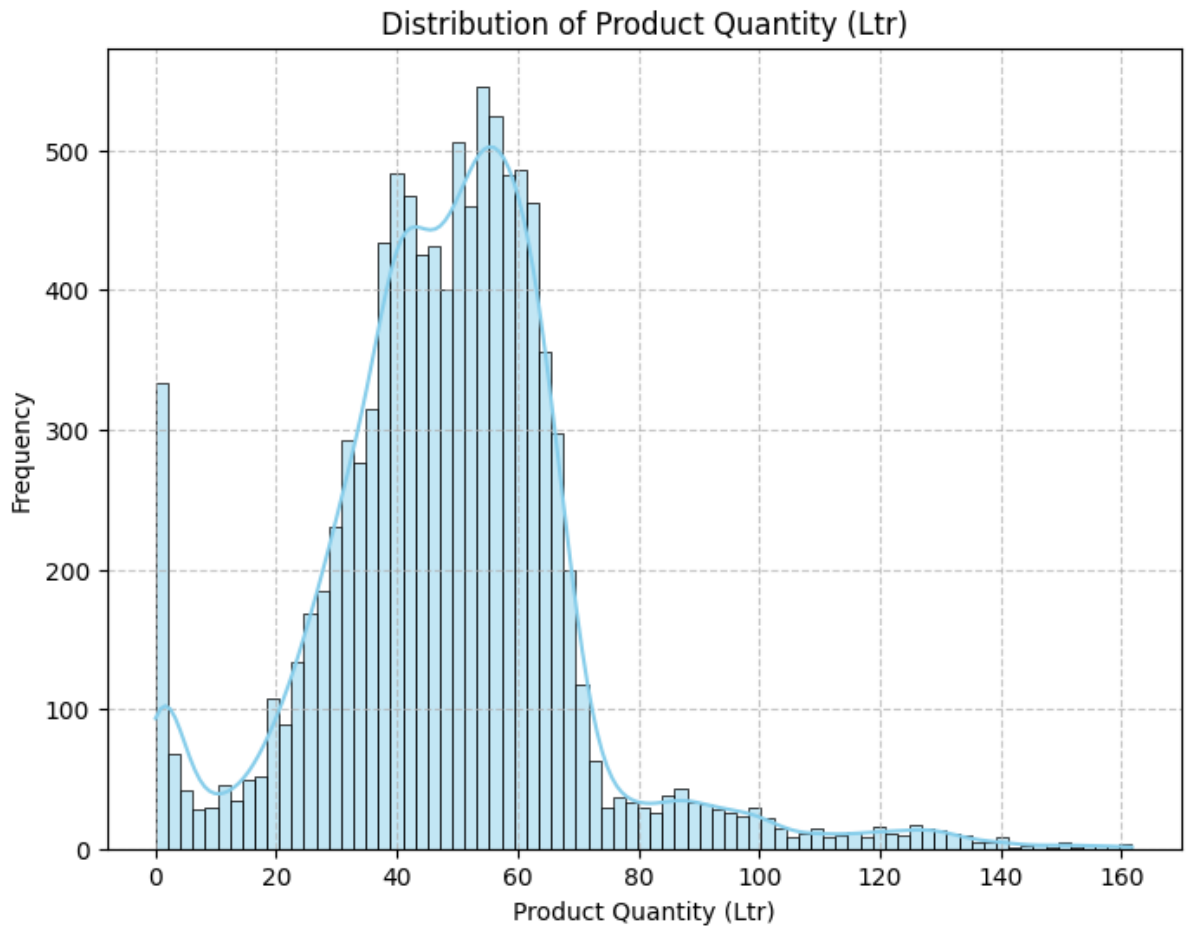


Fig 1.

Distribution of product quantity

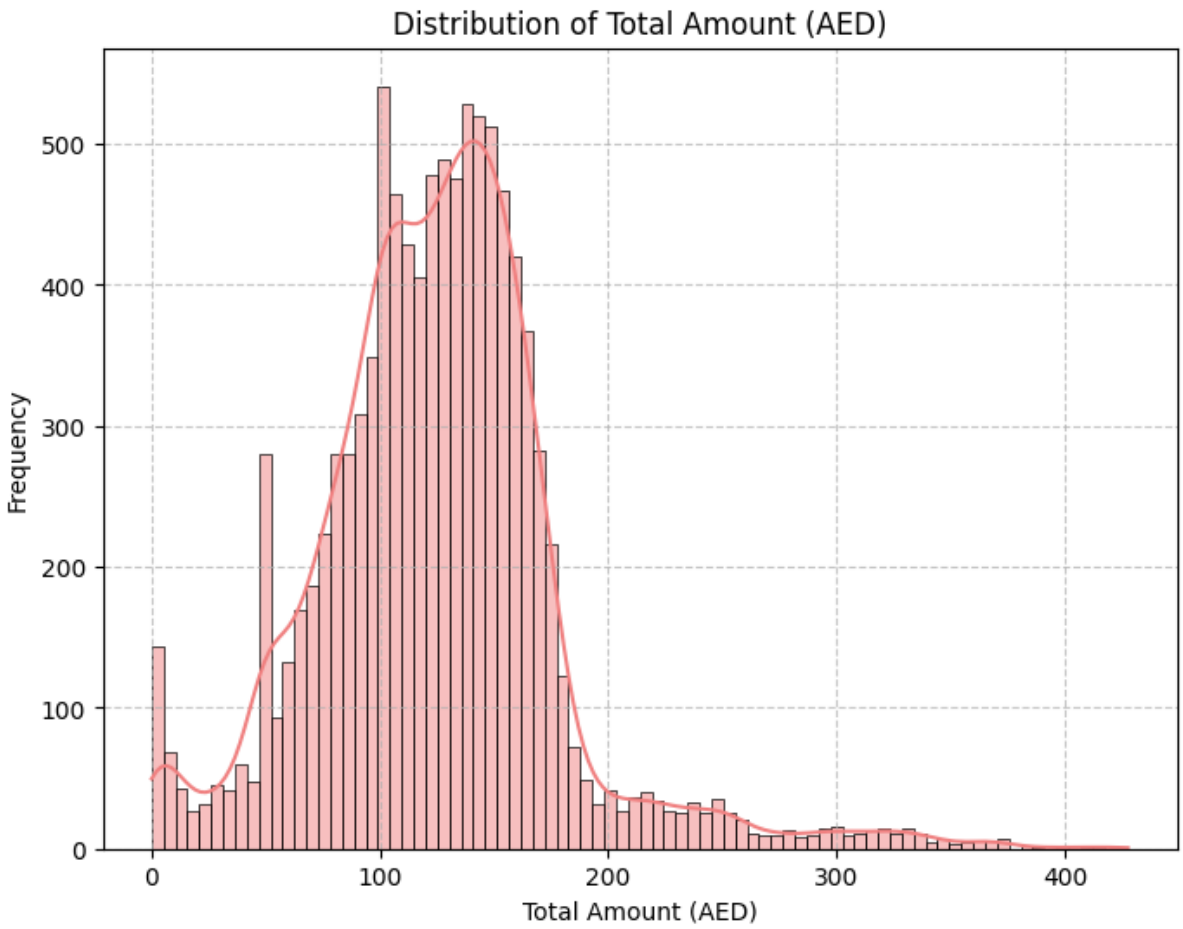


Fig 2.
Distribution of total amount

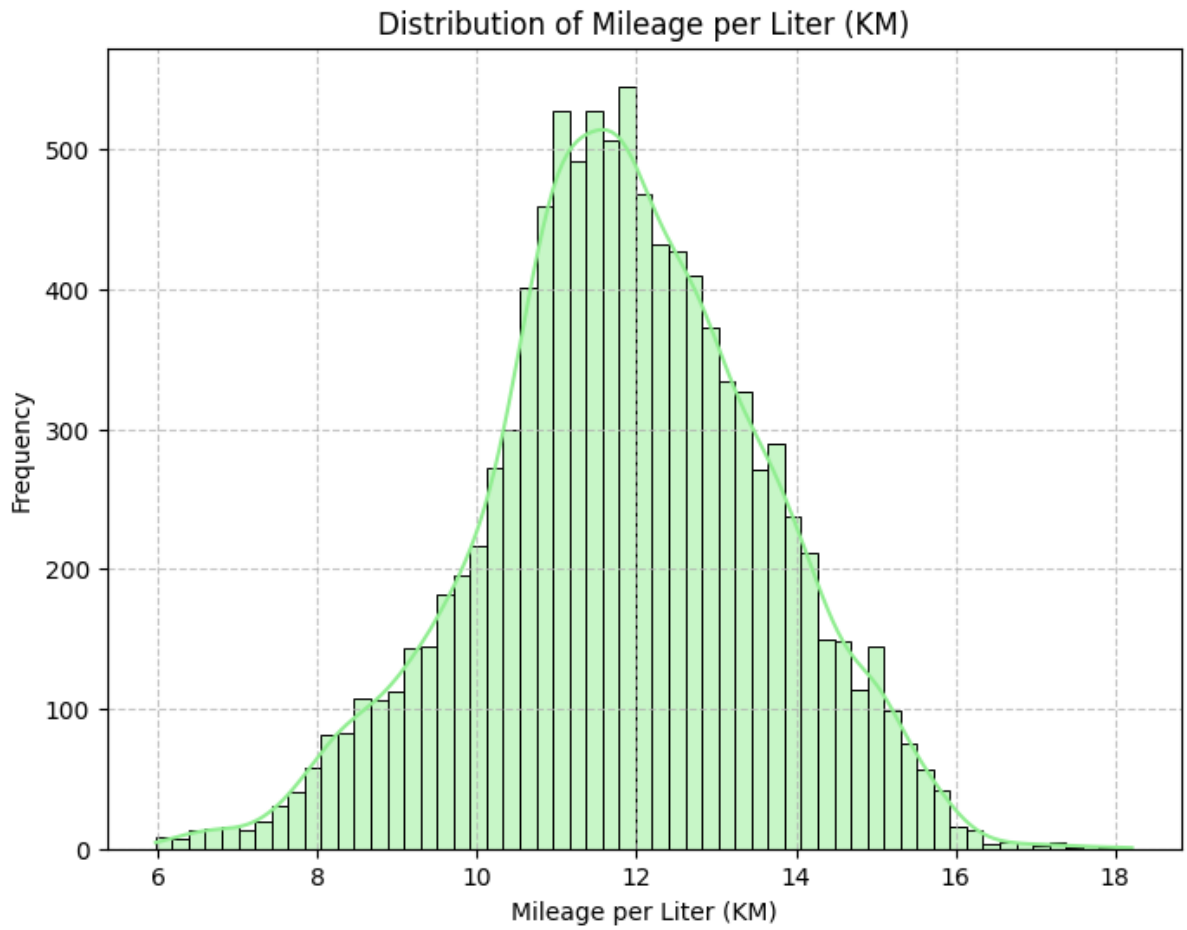


Fig 3.
Distribution of Mileage per liter (KM)

8.2 Relationship analysis

Relationship analysis is used to study how different variables are related to each other. This helps in identifying dependencies and patterns between features.

In this project, scatter plots are used to analyze relationships between:
• Product Quantity and Total Amount
• Mileage and Cost per KM
• Mileage and Efficiency Score

The graphs clearly show that:
• Total amount increases with fuel quantity
• Higher mileage results in lower cost per kilometer

• Efficiency score improves with better mileage values

These relationships are important for building cost prediction and efficiency scoring models.



Fig 4.

Mileage per liter Vs Total amount

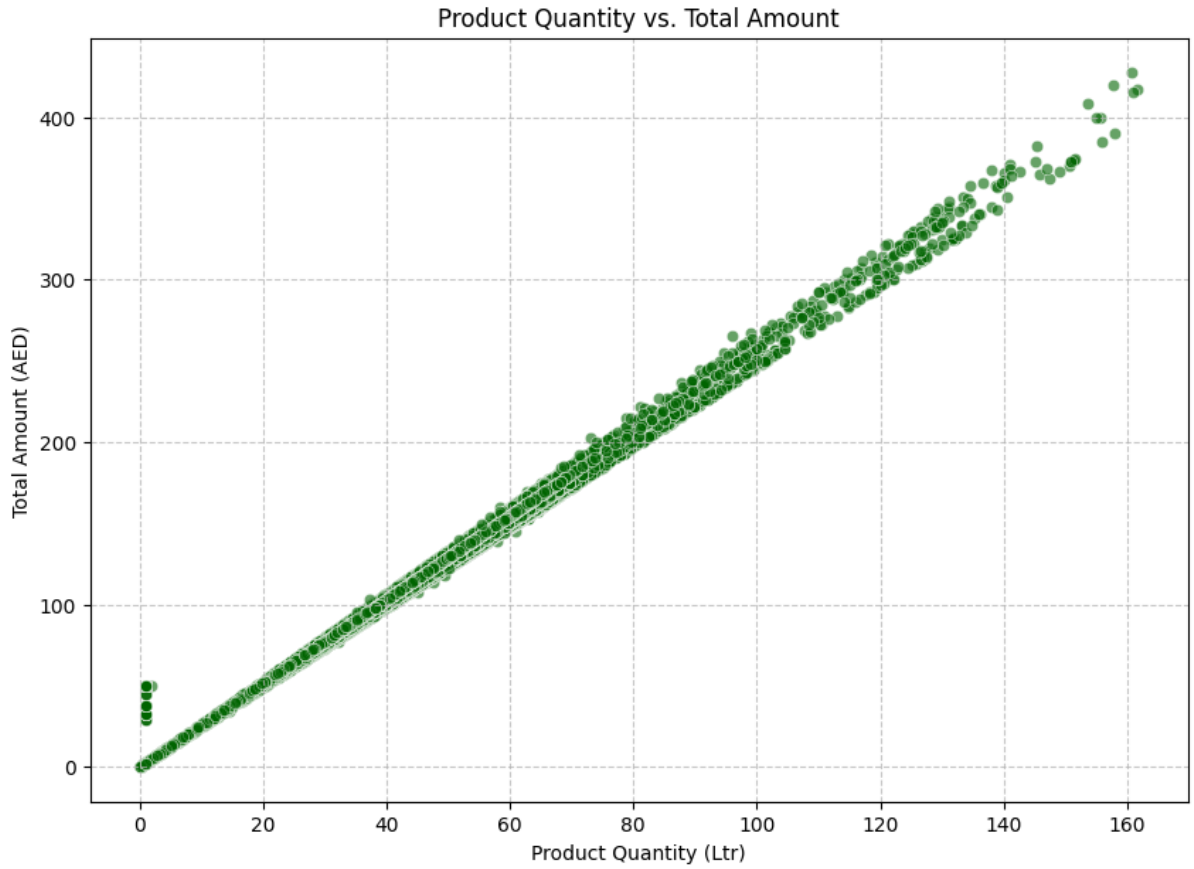


Fig 5.
Product quantity Vs Total amount

8.3 Correlation analysis

Correlation analysis is performed to measure the strength and direction of relationships between numerical variables. A correlation heatmap is used for visualization.

The heatmap shows correlation values ranging from -1 to +1:
• Values close to +1 indicate strong positive correlation
• Values close to -1 indicate strong negative correlation
• Values near 0 indicate weak or no correlation

From the analysis:
• Strong correlation is observed between Product Quantity and Total Amount
• Mileage is strongly related to efficiency-based features
• Derived features such as Cost per KM show meaningful relationships

This helps in selecting important features and removing redundant ones for better model performance.

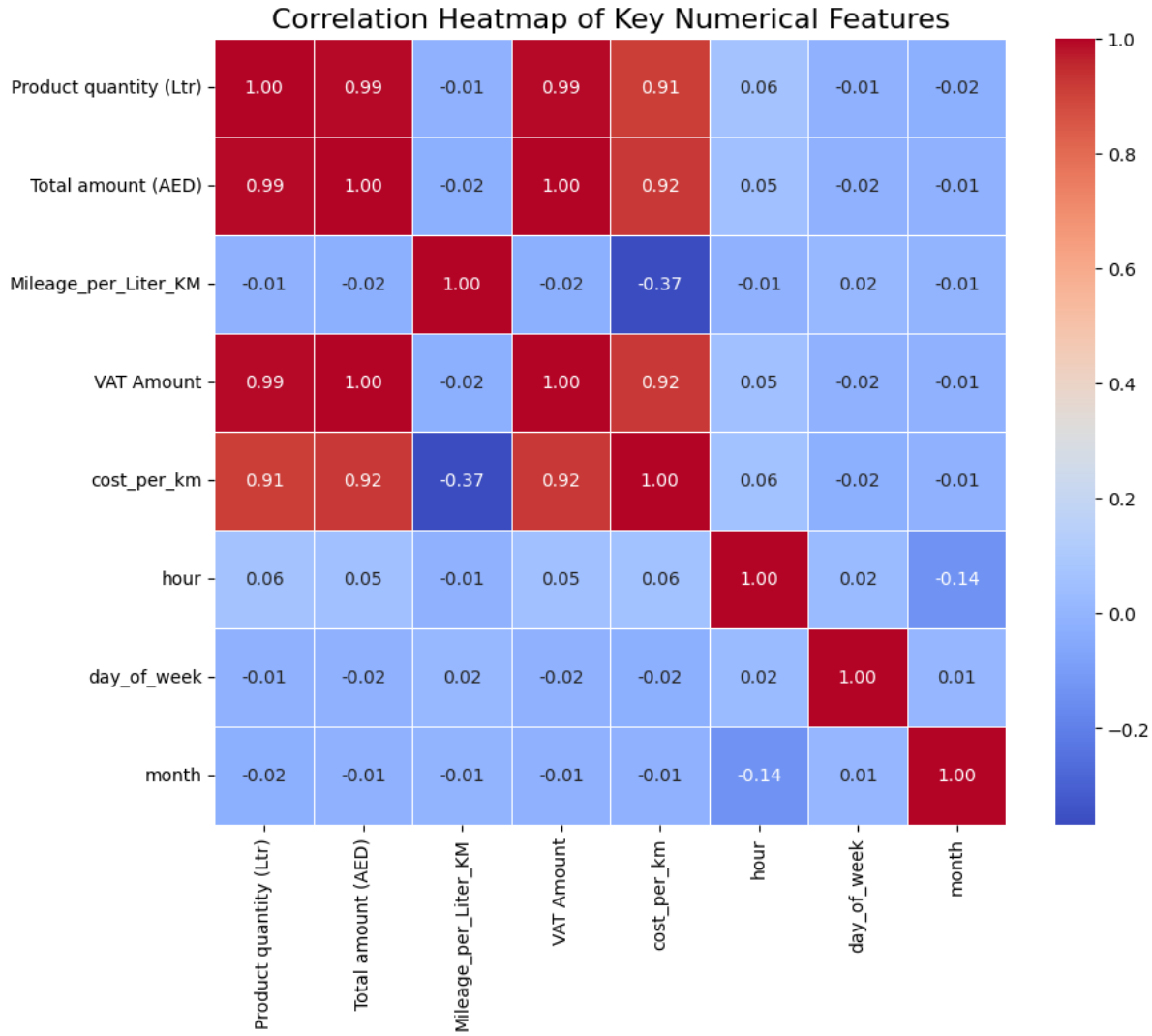


Fig 6.

Correlation Heatmap of Key numerical features

8.4 Outlier detection

Outlier detection is used to identify abnormal data points that deviate significantly from normal values. These may indicate errors or unusual fuel transactions.

In this project, box plots are used to detect outliers in:
• Product Quantity
• Total Amount
• Mileage

From the box plots, it is observed that:
• Some transactions have extremely high fuel quantity or cost

• Certain mileage values are unusually low or high
• These outliers may represent inefficient usage or suspicious activities

Outlier detection is important as it directly supports anomaly detection in the machine learning pipeline.

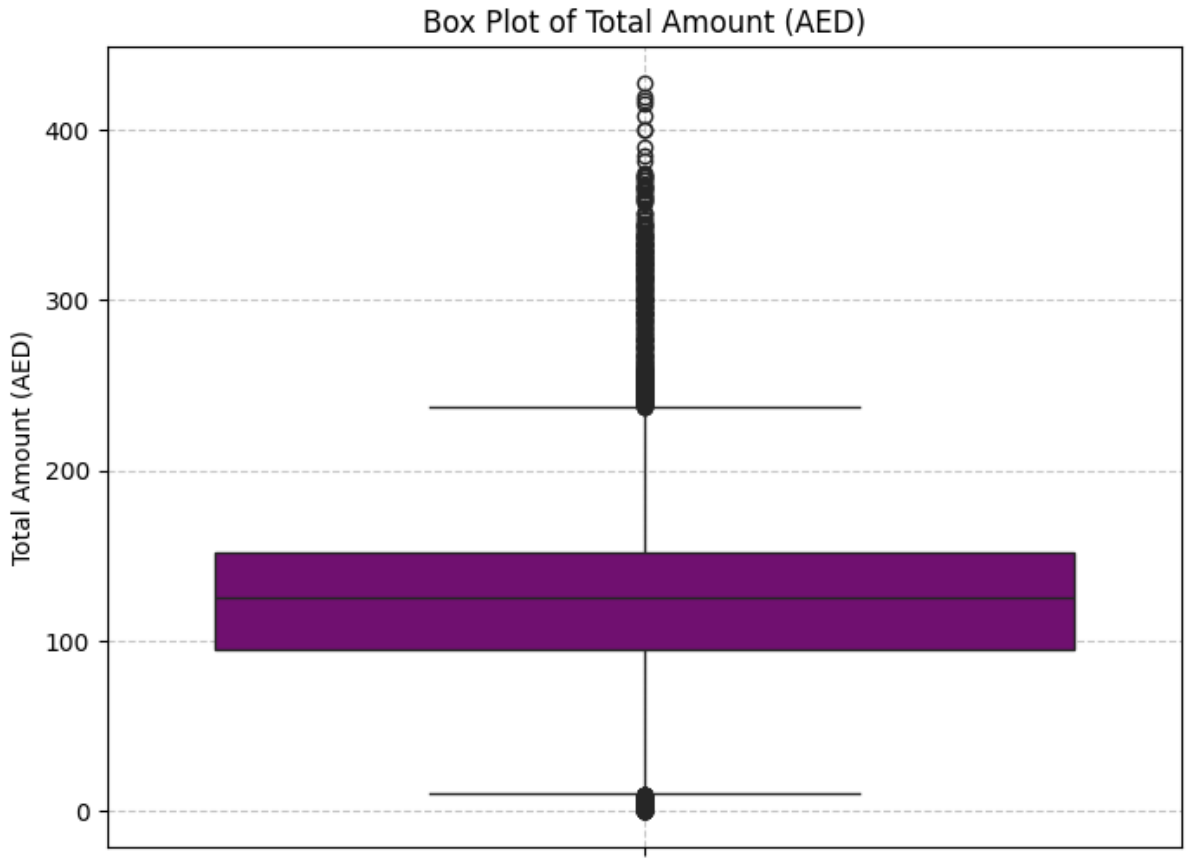


Fig 7.

Box plot of total amount

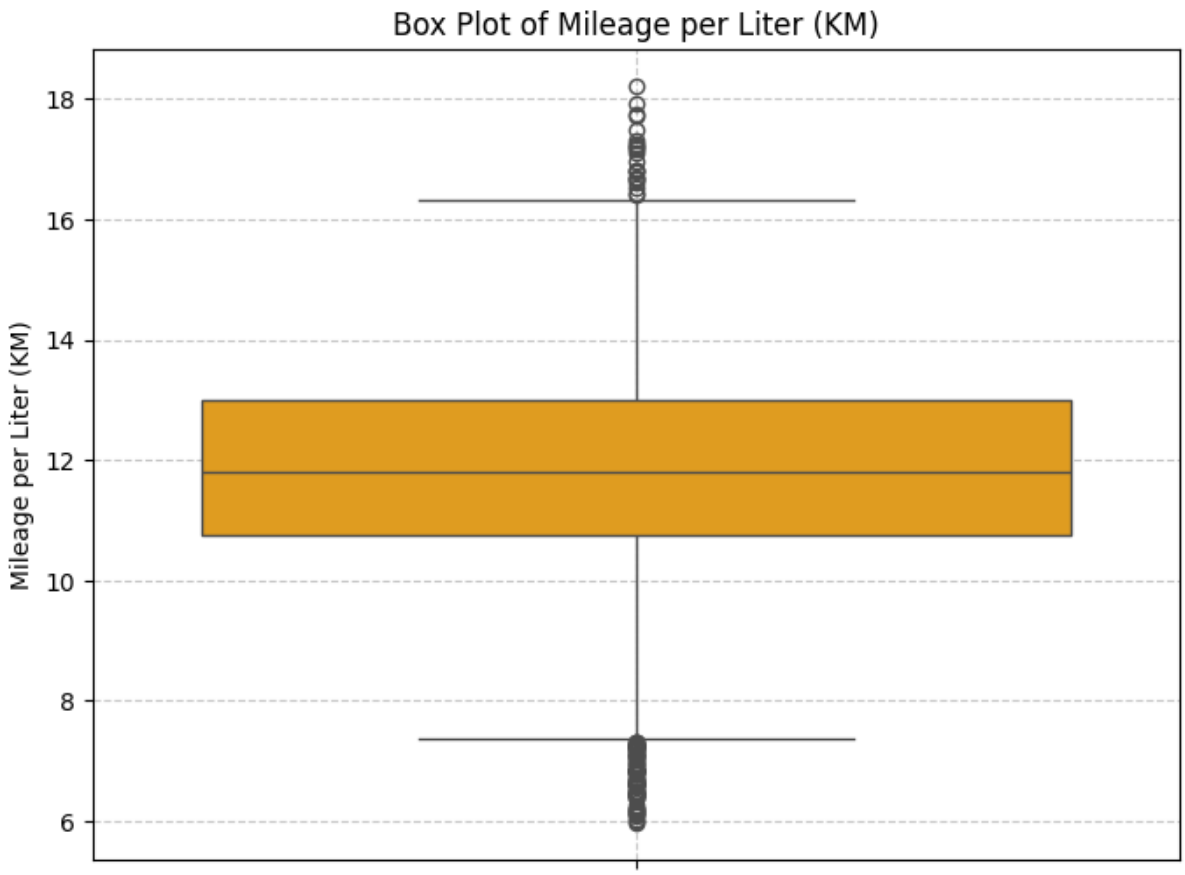


Fig 8.

Box plot of Mileage per liter (KM)

CHAPTER 9: Methodology

9.1 Overview of methodology

The methodology of this project describes the complete process followed to transform raw fuel transaction data into meaningful insights using machine learning techniques. The system follows a structured pipeline that includes data preprocessing, feature engineering, model building, and result generation.

The methodology is designed to support three major functionalities:
• Cost prediction
• Efficiency scoring based on mileage
• Anomaly detection

This pipeline ensures that the system is accurate, scalable, and suitable for real-world applications.

9.2 Workflow of the system

The workflow of the system represents the step-by-step process used in the project.

1. **Data Collection**
   Fuel transaction data is collected from the dataset, including product quantity, total amount, mileage, and transaction details.
2. **Data Preprocessing**
   The data is cleaned by handling missing values, removing duplicates, and converting data into a suitable format.
3. **Feature Engineering**
   New features such as **Cost per KM** and **Efficiency Score** are created using mileage and fuel data.
4. **Model Building**
   Three models are developed:
   • Regression model for cost prediction
   • Scoring mechanism for efficiency evaluation
   • Anomaly detection model for identifying unusual transactions
5. **Model Evaluation**
   The models are evaluated using appropriate performance metrics.
6. **Deployment (Dashboard)**
   The models are integrated into a Streamlit-based dashboard for real-time predictions and visualization.

7. **Result Generation**
   The system outputs predicted cost, efficiency score, and anomaly status.

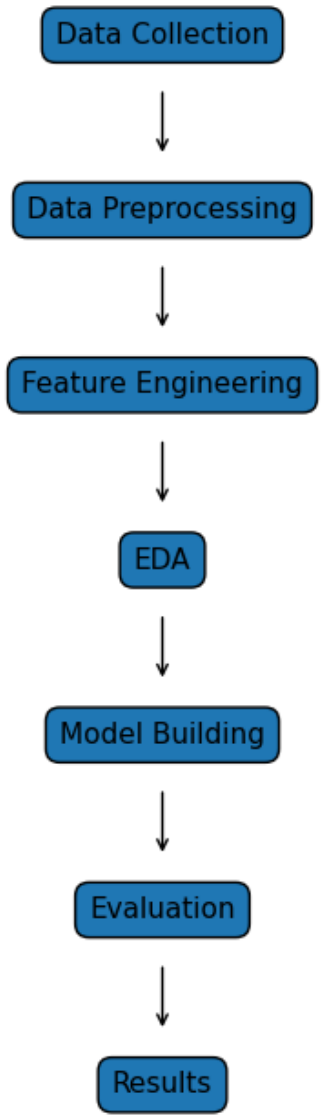


Fig 9.
Workflow of the sysytem

9.3 Data Preprocessing steps

The preprocessing steps followed in this project include:

• Data Cleaning: Removing duplicate and inconsistent records
• Handling Missing Values: Filling missing values using statistical methods
• Encoding: Converting categorical variables into numerical form
• Feature Scaling: Normalizing numerical values
• Outlier Handling: Identifying extreme values

These steps ensure that the data is properly prepared for machine learning models.

CHAPTER 10: MACHINE LEARNING MODELS

10.1 Introduction to machine learning

Machine Learning is a branch of artificial intelligence that enables systems to learn from data and make predictions without explicit programming. In this project, machine learning models are used to analyze fuel transaction data and generate intelligent outputs such as cost prediction, efficiency scoring, and anomaly detection.

10.2 Cost Prediction Model (Regression)

The cost prediction model is used to estimate the total fuel transaction amount based on input features.

The model takes inputs such as:
• Product Quantity
• Mileage
• Cost per KM
• Time-based features

A regression algorithm (such as Random Forest Regression) is used to predict the total transaction cost.

Working:
• The model is trained on historical data
• It learns relationships between fuel quantity, mileage, and cost
• It predicts cost for new input values

Application:
• Predicting fuel expenses
• Supporting cost planning and budgeting

10.3 Efficiency Score Model

The efficiently fuel is used based on mileage and fuel consumption.

The efficiency score is calculated on a scale (e.g., 0–100), where:
• Higher score = better efficiency
• Lower score = poor efficiency

Key factors:

• Mileage (primary factor)
• Fuel consumption
• Cost per KM

Working:
• Features are normalized
• A scoring logic or model assigns efficiency values
• Results help compare vehicle performance

Application:
• Identifying efficient vs inefficient usage
• Improving fleet performance

10.4 Anomaly Detection Model

The anomaly detection model is used to identify unusual or suspicious fuel transactions.

This is implemented using a classification-based or Isolation Forest approach.

Working:
• The model learns normal transaction patterns
• It assigns anomaly scores to new data
• Transactions with abnormal patterns are flagged

Examples of anomalies:
• Very high fuel cost
• Abnormal mileage values
• Irregular transaction behavior

Application:
• Detecting fuel theft or leakage
• Identifying system errors

CHAPTER 11: MODEL EVALUATION

11.1 Evaluation metrics

Model evaluation is performed to measure the performance and accuracy of the machine learning models.

For the **cost prediction model (regression)**, the following metrics are used:

• **Mean Absolute Error (MAE)**
Measures the average difference between predicted and actual values
• **Mean Squared Error (MSE)**
Measures squared differences and penalizes large errors
• **Root Mean Squared Error (RMSE)**
Provides error in the same unit as output
• **R-squared (R² Score)**
Measures how well the model explains data variability
For **efficiency scoring**, evaluation is based on:
• Logical correctness of scoring
• Consistency with mileage values
• Interpretability of results
For **anomaly detection**, evaluation is based on:
• Accuracy of identifying unusual transactions
• Comparison with observed abnormal patterns
• Model confidence scores

11.2 Performance Analysis

Cost Prediction Model:
The regression model provides accurate predictions of fuel transaction cost. The error values (MAE, RMSE) are low, indicating good performance. The model successfully captures relationships between fuel quantity, mileage, and cost.

Efficiency Scoring Model:
The efficiency scoring system effectively evaluates fuel usage performance. Transactions with higher mileage receive higher scores, while inefficient ones are assigned lower scores. This helps in identifying performance differences across vehicles.

Anomaly Detection Model:

The anomaly detection model successfully identifies unusual transactions. It detects abnormal patterns such as high cost or irregular mileage values. These results help in improving system monitoring and security.

Overall System Performance:

The combination of prediction, scoring, and anomaly detection models provides a complete solution for fuel analysis. The system performs efficiently and generates meaningful insights that support decision-making and operational improvement.

CHAPTER 12: SYSTEM ARCHITECTURE

12.1 Architecture diagram

The system architecture represents the overall structure of the Fleet Fuel Intelligence system and shows how different components interact with each other. It provides a clear understanding of how data flows through the system and how machine learning models are integrated into a unified pipeline.

The architecture of this project is designed as a multi-stage pipeline consisting of the following components:

• **Data Input Layer**
This layer includes the dataset containing fuel transaction details such as product quantity, total amount, mileage, and transaction time.

• **Data Preprocessing Layer**
In this stage, raw data is cleaned and prepared. Missing values are handled, duplicates are removed, and data is transformed into a suitable format for analysis.

• **Feature Engineering Layer**
Important features are selected, and new features such as **Cost per KM** and **Efficiency Score** are created using mileage and fuel data.

• **Machine Learning Layer**
This layer includes three main components:
1.Cost Prediction Model (Regression)
2.Efficiency Scoring Model
3. Anomaly Detection Model

Each model performs a specific task and contributes to the overall system functionality.

• **Output Layer**
The final outputs include predicted cost, efficiency score, and anomaly status. These outputs are displayed to the user through an interactive

dashboard.

The architecture follows a modular design, allowing easy scalability and integration with real-time systems.

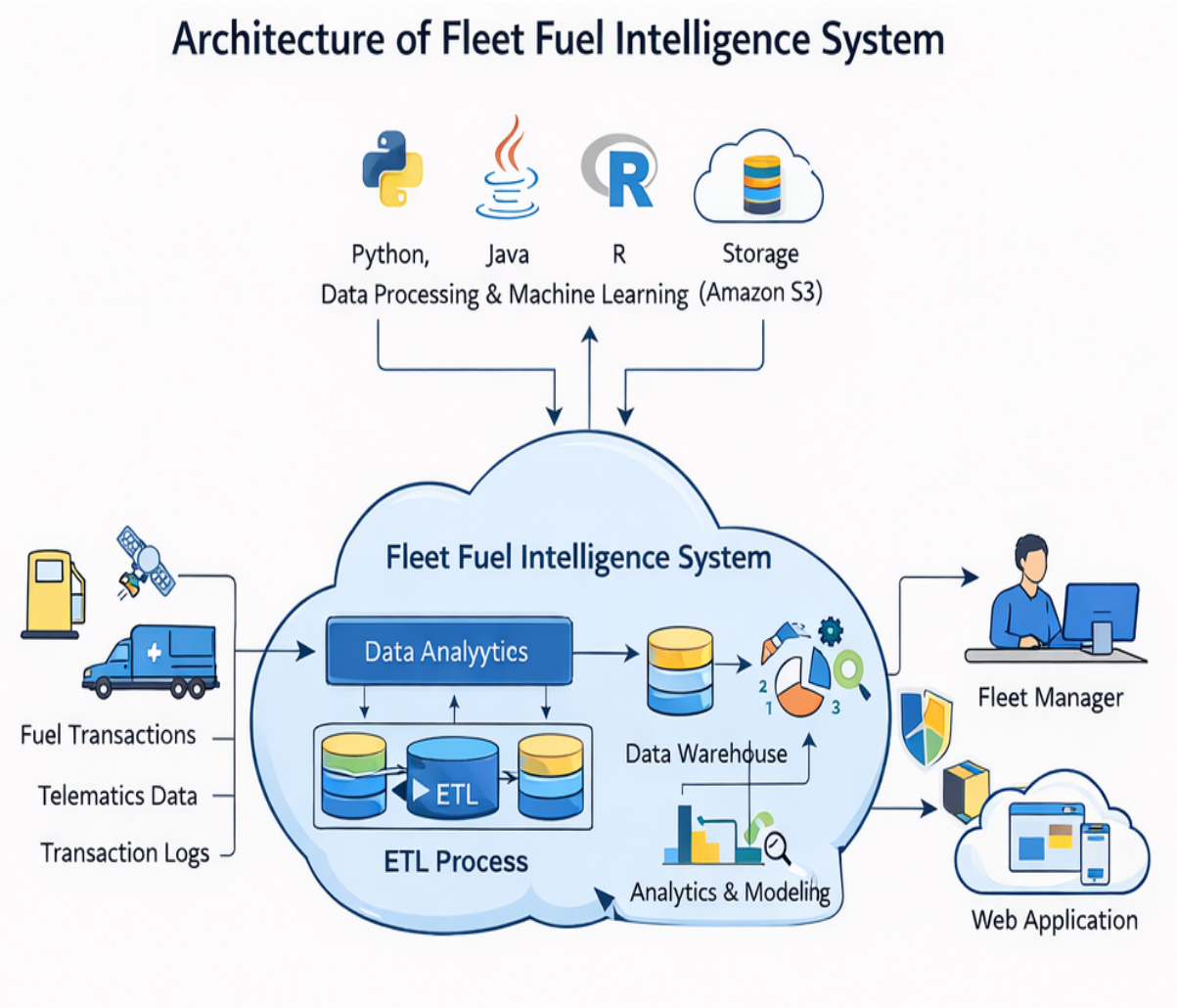


Fig 10.

Architecture Diagram

12.2 Working flow

The working flow of the system describes how data is processed step-by-step:

1. The system takes fuel transaction data as input
2. The data is preprocessed to remove errors and inconsistencies
3. Feature engineering is applied to generate new useful features
4. The processed data is passed to machine learning models
5. The cost prediction model estimates transaction cost
6. The efficiency model calculates efficiency score
7. The anomaly detection model identifies unusual transactions
8. Results are displayed on the dashboard

This structured workflow ensures efficient processing and accurate results.

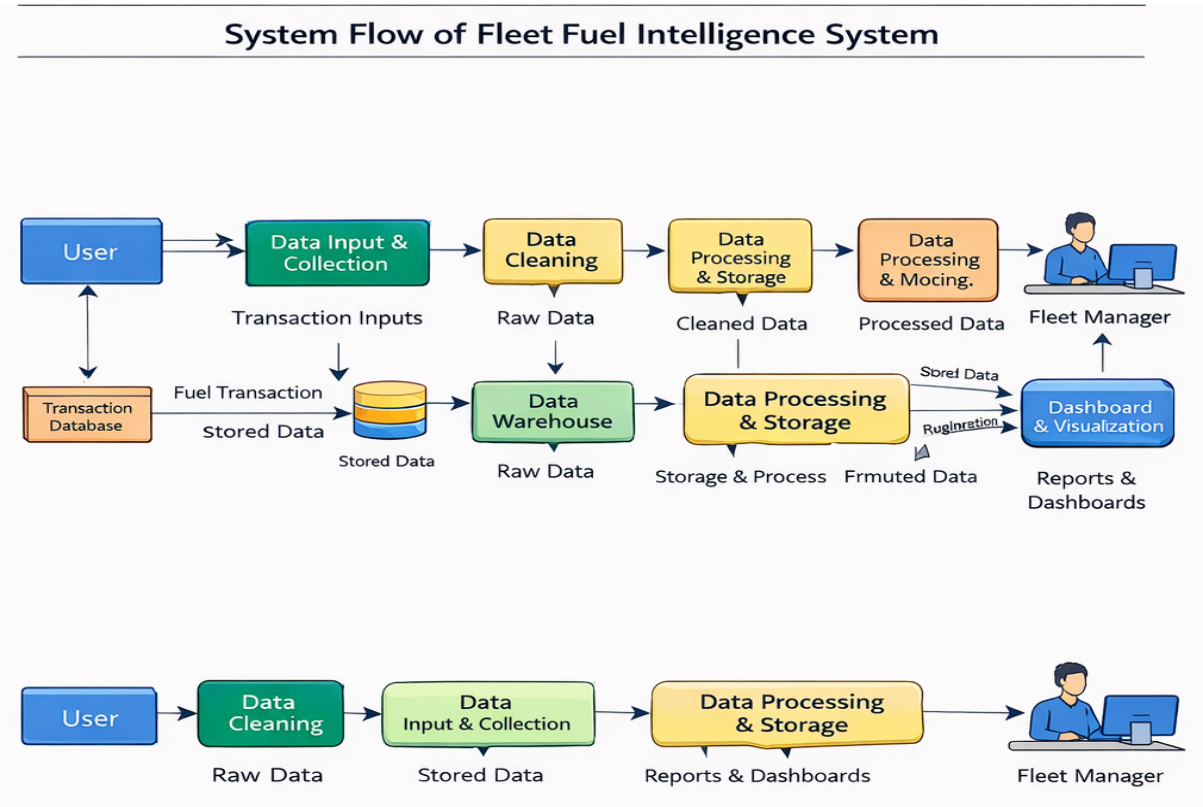


Fig 11.

Workflow Diagram

CHAPTER 13: DATA FLOW DIAGRAMS (DFD)

13.1 DFD Level 0 Fleet fuel intelligence system (context diagram)

The Level 0 Data Flow Diagram represents the overall system as a single process. It shows the interaction between the user and the system.

In this system:
• The user provides fuel transaction input
• The system processes the data using machine learning models
• The system generates outputs such as predicted cost, efficiency score, and anomaly status

This diagram provides a high-level overview of the system.

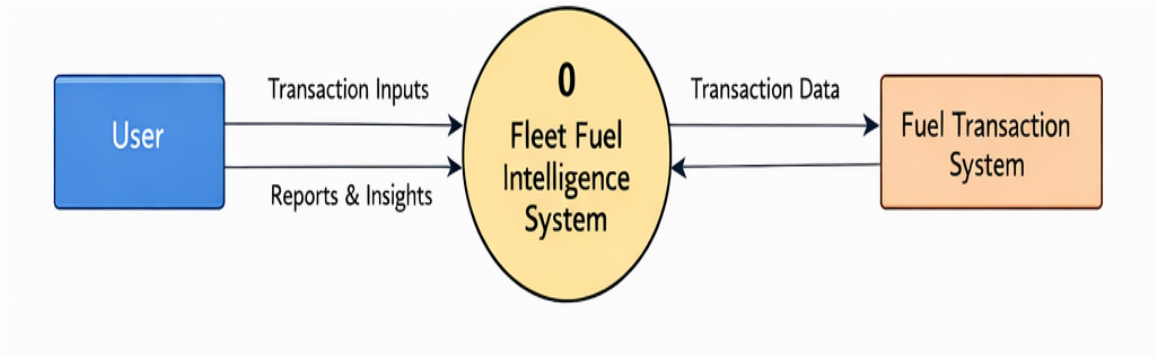


Fig 12.

DFD Level 0

13.2 DFD Level 1 Top level Diagram

The Level 1 DFD breaks the system into major processes:

- Data Input Module
- Data Preprocessing Module
- Feature Engineering Module
- Machine Learning Module
- Output Module

Each module performs a specific function and passes data to the next stage. This level shows how data flows between different components of the system.

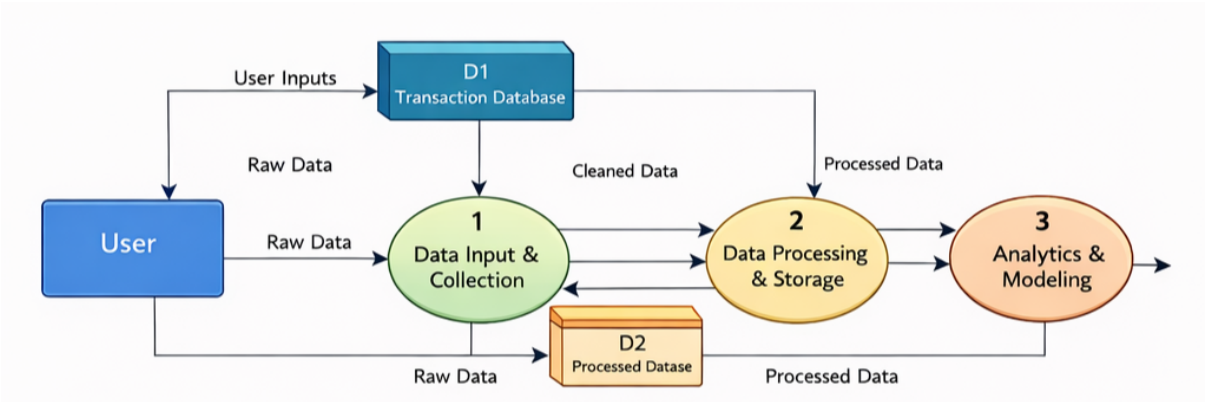


Fig 13.

DFD Level 1

13.3 DFD Level 2 Detailed Diagram

The Level 2 DFD provides a detailed view of internal processes within each module.

It includes:
• Data cleaning and transformation steps
• Feature creation such as cost per KM and efficiency score
• Model operations for prediction, scoring, and anomaly detection
• Output generation process

This level explains how the system processes data internally in detail.

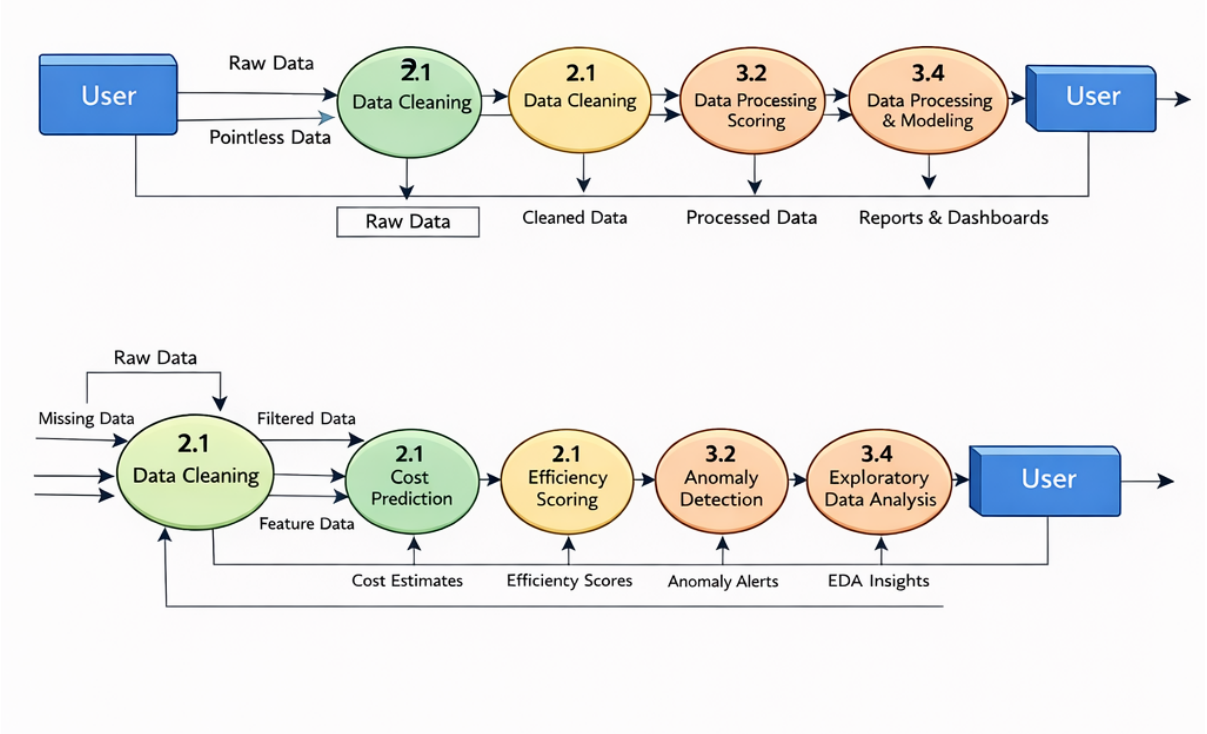


Fig 14.

DFD Level 2

CHAPTER 14: ENTITY RELATIONSHIP DIAGRAM(ER DIAGRAM)

14.1 ER diagram description

The Entity Relationship (ER) Diagram represents the logical structure of the database used in the system. It shows how different entities are connected and how data is organized.

In this project, the **Transaction** entity acts as the central component connecting all other entities. Each transaction stores detailed information about fuel usage and related attributes.

Entities and attributes:

The system consists of the following main entities:
1. Transaction
This is the central entity that stores details of each fuel transaction.
Attributes:
• transaction_id (Primary Key)
• product_quantity
• total_amount
• mileage
• hour
• day

• month
2. Vehicle
Stores information about vehicles involved in transactions.
Attributes:
• vehicle_id (Primary Key)
• vehicle_info
3. Station
Represents fuel stations where transactions occur.
Attributes:
• station_id (Primary Key)
• station_name
4. Product
Stores fuel type details.
Attributes:
• product_id (Primary Key)
• product_name
5. Payment
Stores payment-related information.
Attributes:
• payment_id (Primary Key)
• payment_method
6. Beneficiary
Represents the user or organization receiving the service.
Attributes:
• beneficiary_id (Primary Key)
• beneficiary_name

Relationships Between Entities

• Vehicle → Transaction (One-to-Many)
• Station → Transaction (One-to-Many)
• Product → Transaction (One-to-Many)
• Payment → Transaction (One-to-Many)
• Beneficiary → Transaction (One-to-Many)

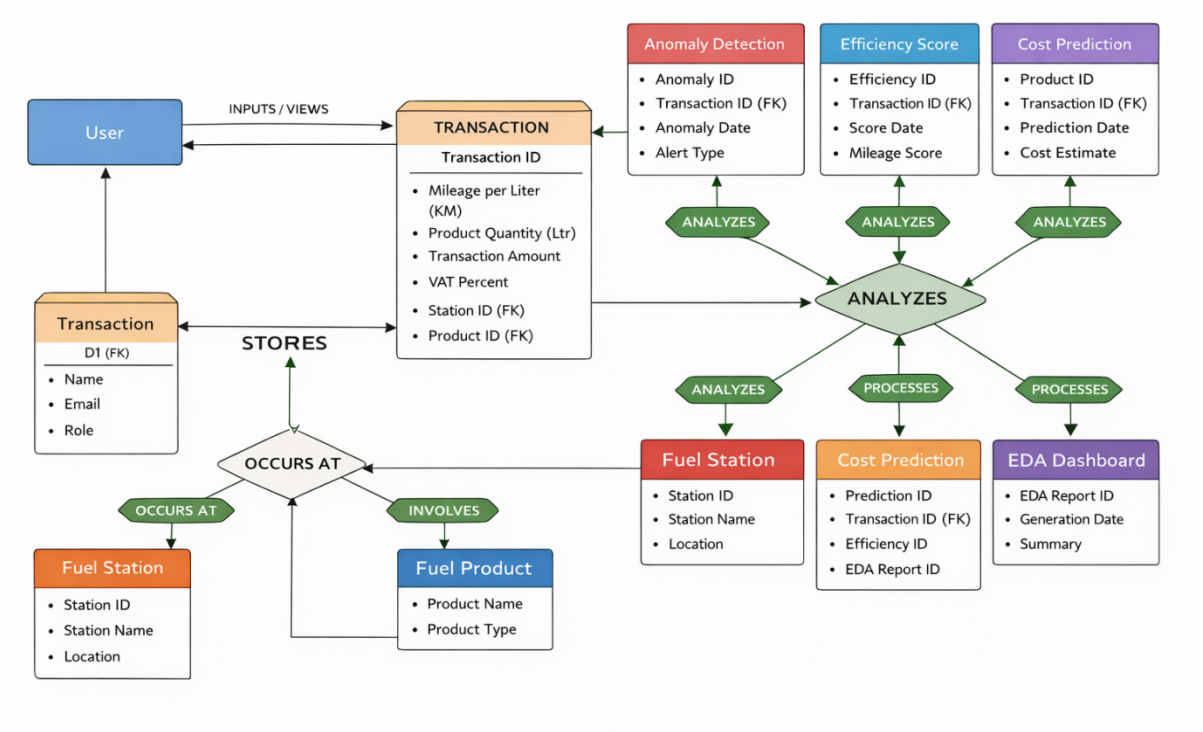


Fig 15.
ER Diagram

Working of the ER Diagram

The ER diagram shows that all entities are connected through the Transaction entity. When a fuel transaction occurs:

- A vehicle performs the transaction
- Fuel is filled at a station
- A specific product is used
- Payment is made using a method
- A beneficiary is associated

This structured relationship ensures efficient data storage and retrieval for analysis and machine learning operations.

CHAPTER 15: RESULTS AND DISCUSSION

15.1 Cost Prediction results

The cost prediction model was implemented using a regression-based machine learning approach. The model was trained on historical fuel transaction data, including features such as product quantity, mileage, and time-based attributes.

The model successfully predicts the total transaction cost with good accuracy. The predicted values are close to the actual values, indicating that the model has effectively learned the relationship between fuel quantity, mileage, and cost.

Key observations:
• The model performs well for both small and large transactions

• Prediction accuracy improves with better feature representation
• Mileage plays an important role in improving prediction performance

The results demonstrate that the model can be effectively used for forecasting fuel expenses and supporting budgeting decisions.

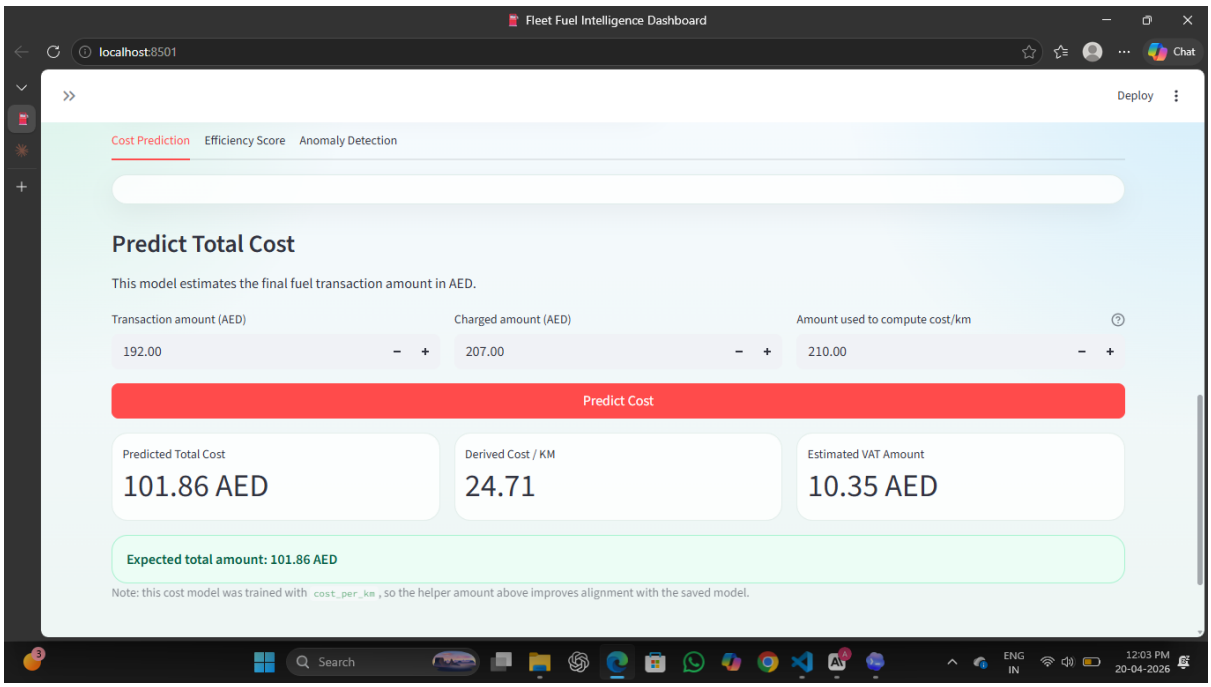


Fig 16.
Cost Prediction results

15.2 Efficiency Scoring Results

The efficiency scoring model evaluates fuel usage performance based on mileage and fuel consumption. The score is generated on a scale of 0 to 100, where higher values indicate better efficiency.

The results show that:
• Transactions with higher mileage receive higher efficiency scores
• Lower mileage results in lower efficiency scores
• The scoring system clearly differentiates between efficient and inefficient usage

This model helps in identifying vehicles or transactions that are not performing efficiently and require attention.

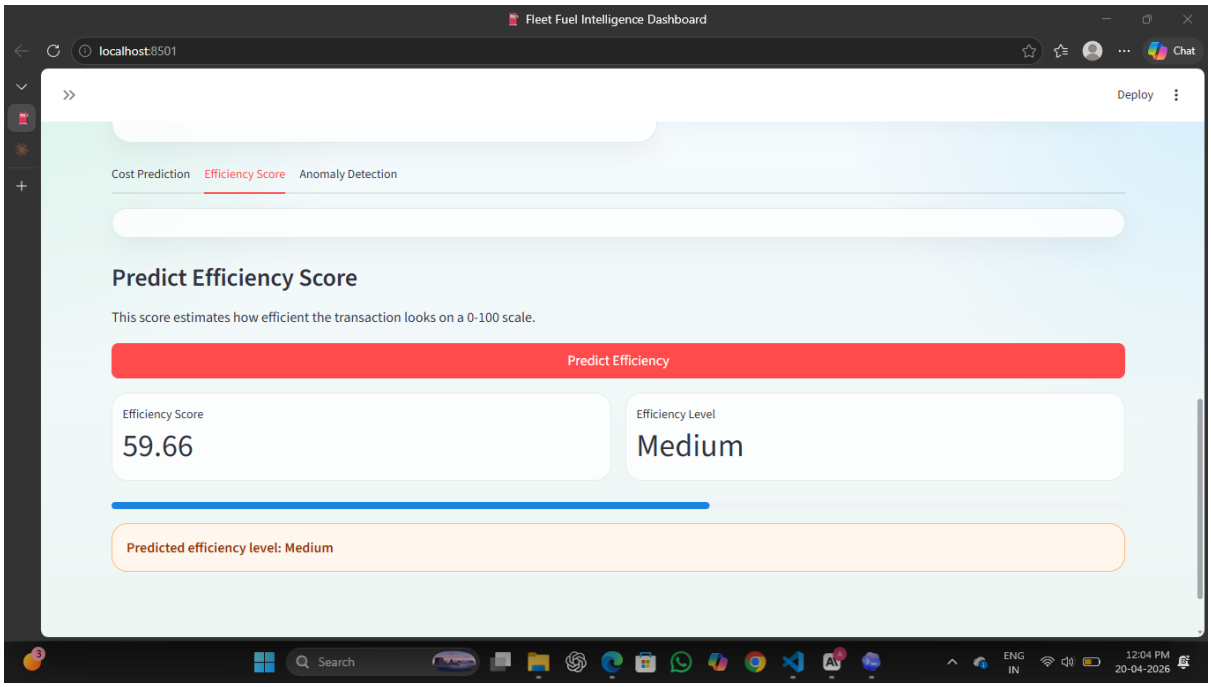


Fig 17.
Efficiency Scoring Results

15.3 Anomaly Detection results

The anomaly detection model is used to identify unusual or suspicious transactions. The model analyzes patterns in fuel consumption, cost, and mileage to detect deviations from normal behavior.

The results indicate that:
• Transactions with abnormal fuel cost are flagged
• Unusual mileage values are detected
• Irregular transaction patterns are identified as anomalies

These anomalies may indicate issues such as fuel wastage, system errors, or potential fraud. The model provides a reliable mechanism for monitoring and controlling such activities.

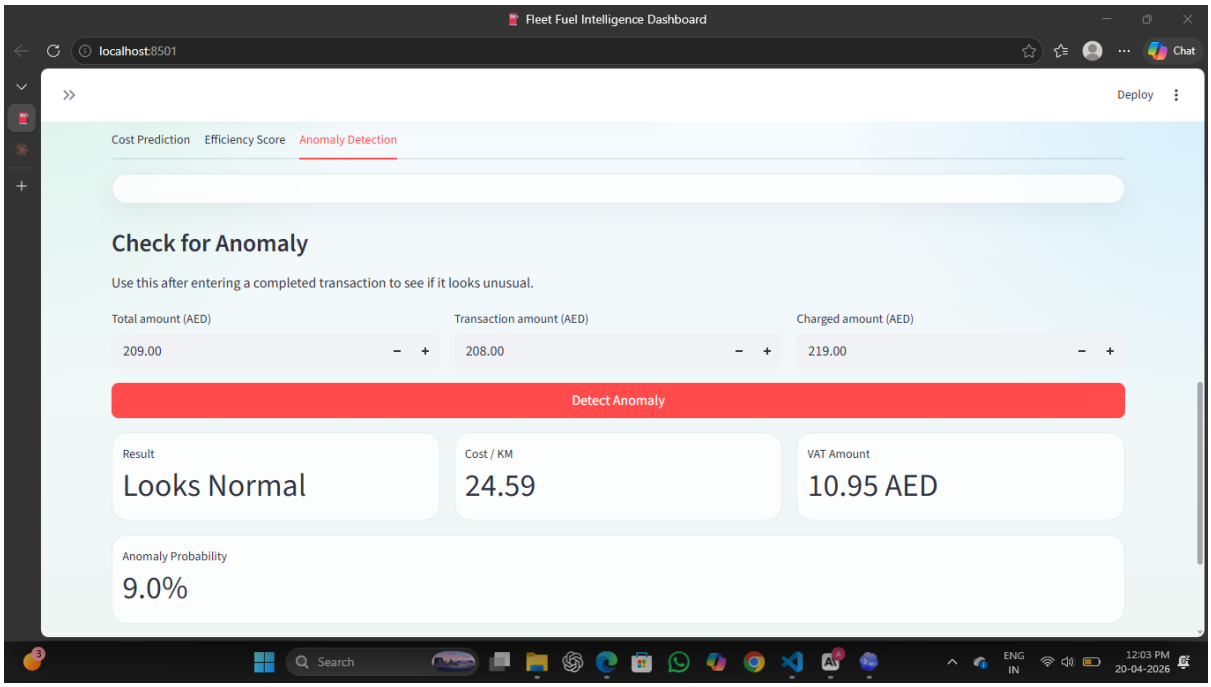


Fig 18.
Anomaly Detection results

15.4 Interpretation of results

The overall results of the system demonstrate the effectiveness of integrating machine learning techniques for fuel data analysis.

From the cost prediction results, it is clear that the system can accurately estimate fuel transaction costs. The efficiency scoring model provides meaningful insights into fuel usage performance, while the anomaly detection model helps in identifying unusual transactions.

The combined output of these models enables:
• Better monitoring of fuel usage
• Identification of inefficient operations
• Detection of suspicious activities
• Data-driven decision-making

Overall, the system proves to be effective, reliable, and suitable for real-world fleet fuel management applications.

CHAPTER 16: SNAPSHOTS

Front end screenshots :-

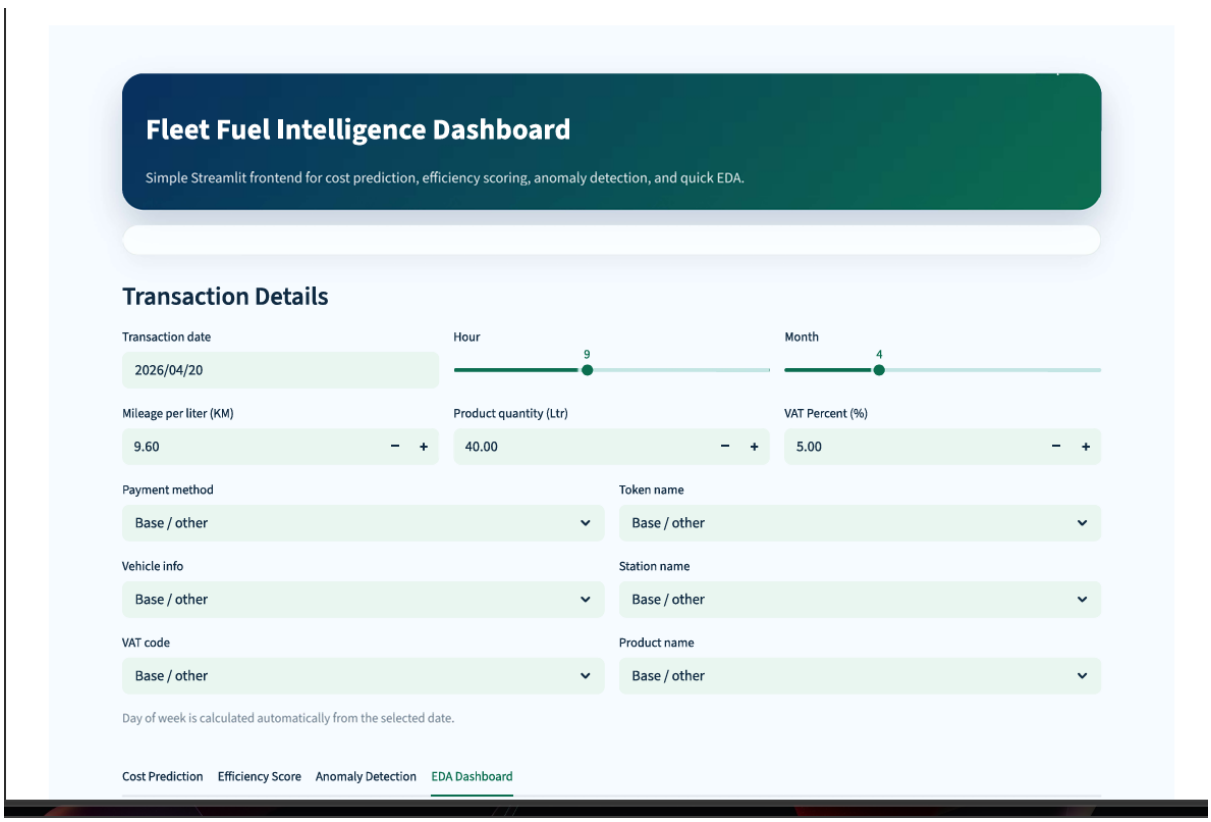

Fleet Fuel Intelligence Dashboard
Simple Streamlit frontend for cost prediction, efficiency scoring, anomaly detection, and quick EDA.
Transaction Details
Transaction date
2026/04/20
Hour
9
Month
4
Mileage per liter (KM)
9.60
Product quantity (Ltr)
40.00
VAT Percent (%)
5.00
Payment method
Base / other
Token name
Base / other
Vehicle info
Base / other
Station name
Base / other
VAT code
Base / other
Product name
Base / other
Day of week is calculated automatically from the selected date.
Cost Prediction
Efficiency Score
Anomaly Detection
EDA Dashboard

Fig 19.

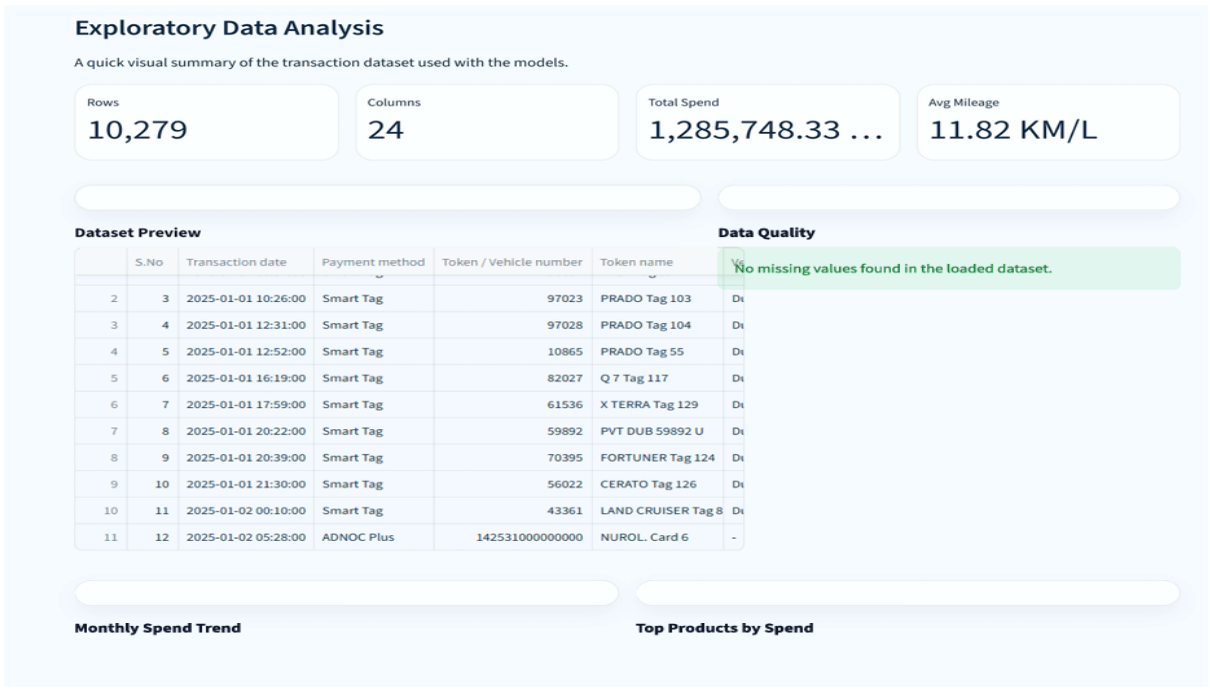

Exploratory Data Analysis
A quick visual summary of the transaction dataset used with the models.
Rows
10,279
Columns
24
Total Spend
1,285,748.33 ...
Avg Mileage
11.82 KM/L
Dataset Preview
Data Quality
No missing values found in the loaded dataset.
S.No
Transaction date
Payment method
Token / Vehicle number
Token name
2 3 2025-01-01 10:26:00 Smart Tag 97023 PRADO Tag 103
3 4 2025-01-01 12:31:00 Smart Tag 97028 PRADO Tag 104
4 5 2025-01-01 12:52:00 Smart Tag 10865 PRADO Tag 55
5 6 2025-01-01 16:19:00 Smart Tag 82027 Q 7 Tag 117
6 7 2025-01-01 17:59:00 Smart Tag 61536 X TERRA Tag 129
7 8 2025-01-01 20:22:00 Smart Tag 59892 PVT DUB 59892 U
8 9 2025-01-01 20:39:00 Smart Tag 70395 FORTUNER Tag 124
9 10 2025-01-01 21:30:00 Smart Tag 56022 CERATO Tag 126
10 11 2025-01-02 00:10:00 Smart Tag 43361 LAND CRUISER Tag 8
11 12 2025-01-02 05:28:00 ADNOC Plus 142531000000000 NUROL. Card 6
Monthly Spend Trend
Top Products by Spend

Fig 20.

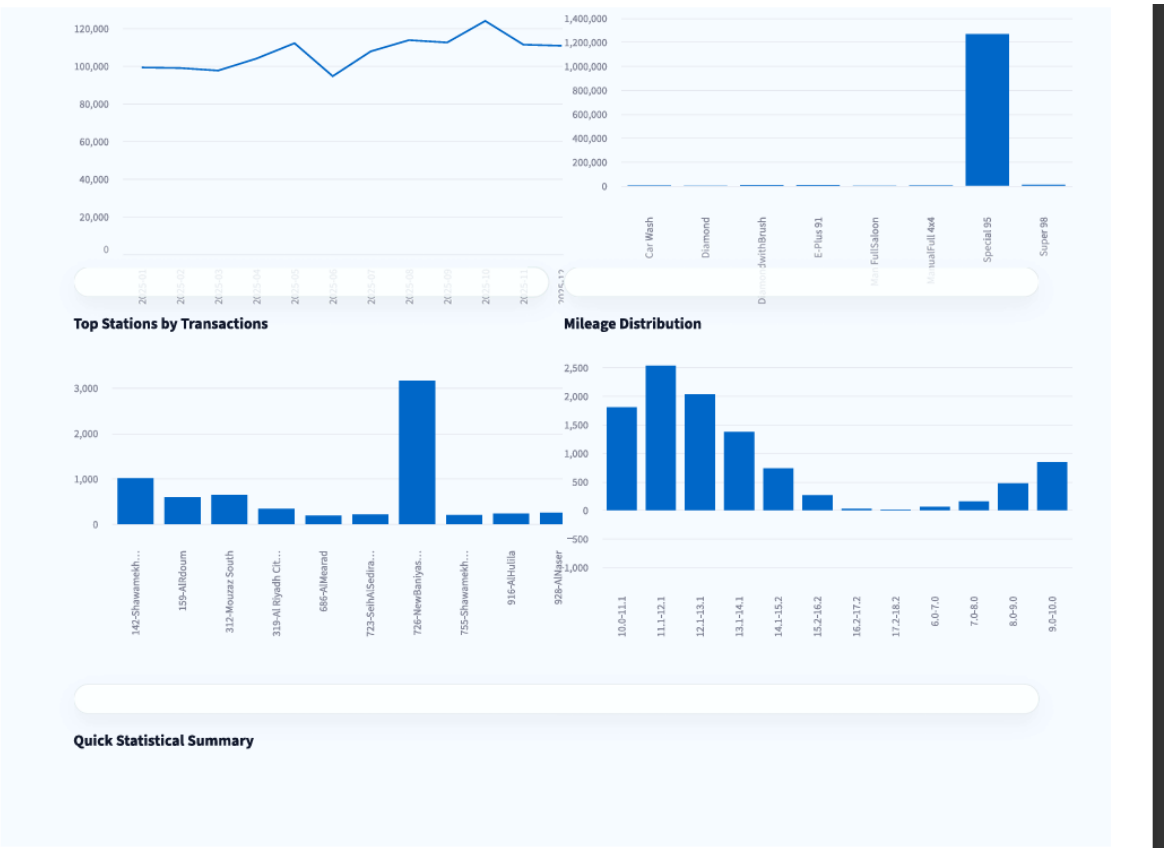

Top Stations by Transactions
Mileage Distribution
Quick Statistical Summary

Fig 21.

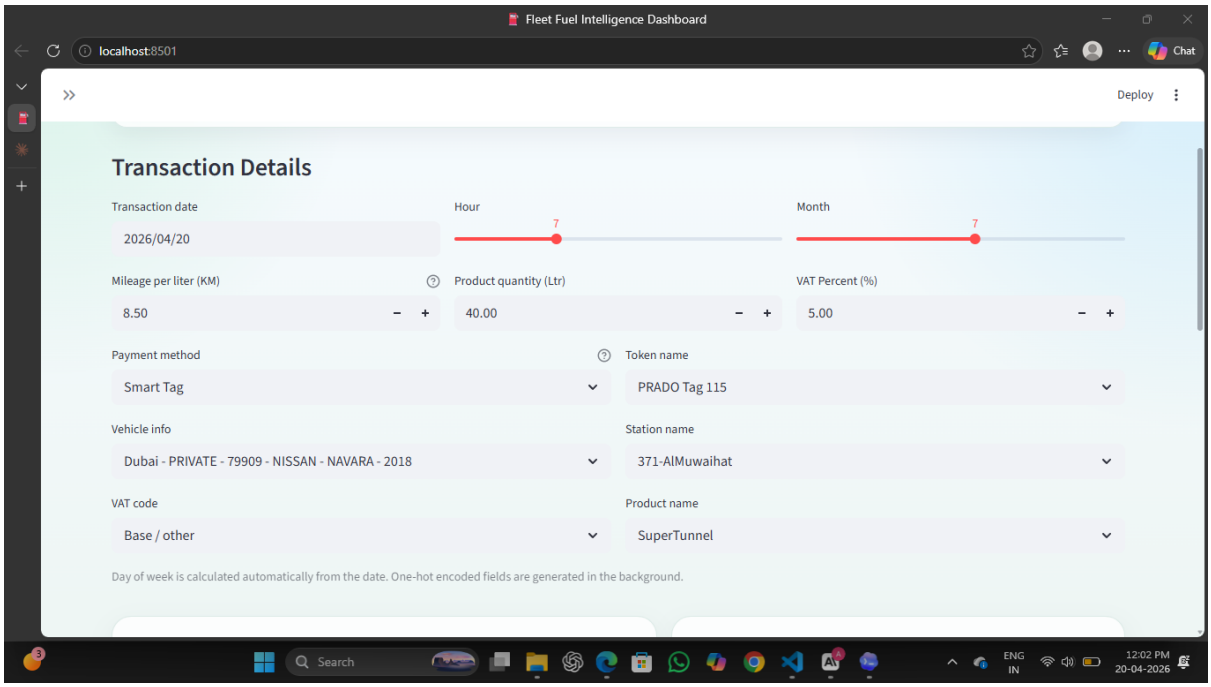

Fleet Fuel Intelligence Dashboard
localhost:8501
Chat
Deploy
Transaction Details
Transaction date
2026/04/20
Hour
7
Month
7
Mileage per liter (KM)
8.50
Product quantity (Ltr)
40.00
VAT Percent (%)
5.00
Payment method
Smart Tag
Token name
PRADO Tag 115
Vehicle info
Dubai - PRIVATE - 79909 - NISSAN - NAVARA - 2018
Station name
371-AlMuwaihat
VAT code
Base / other
Product name
SuperTunnel
Day of week is calculated automatically from the date. One-hot encoded fields are generated in the background.
Search
ENG
IN
12:02 PM
20-04-2026

Fig 22.

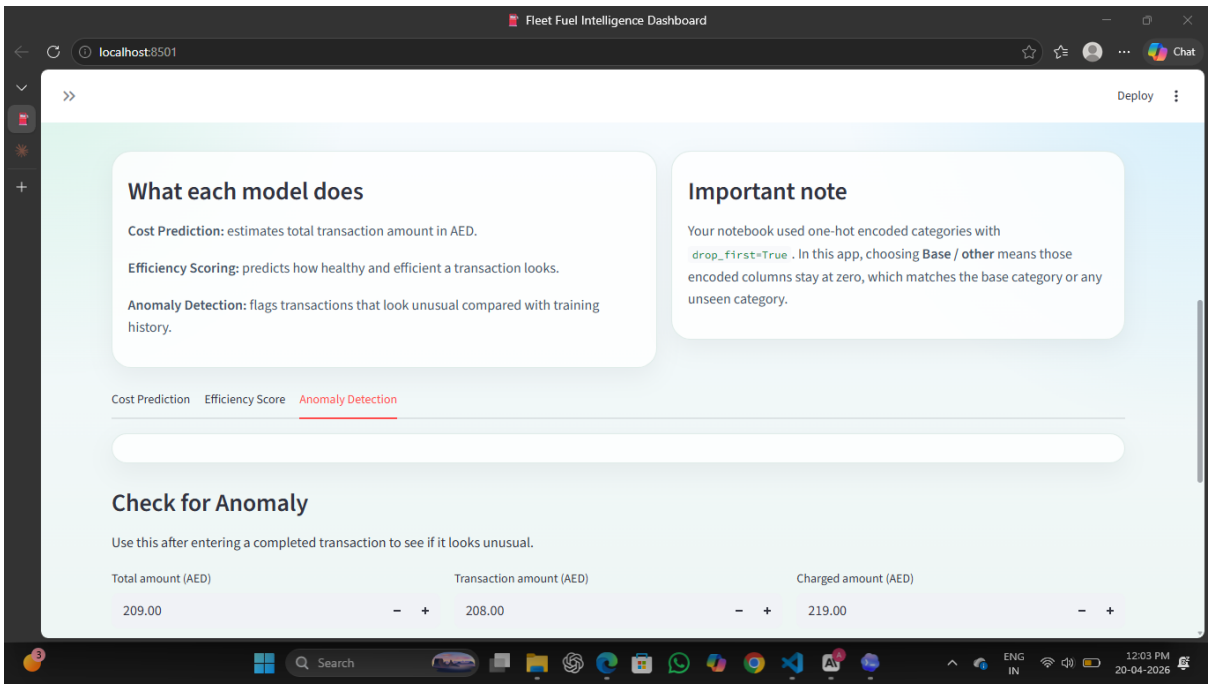

Fleet Fuel Intelligence Dashboard
localhost:8501
Chat
Deploy
What each model does
Cost Prediction: estimates total transaction amount in AED.
Efficiency Scoring: predicts how healthy and efficient a transaction looks.
Anomaly Detection: flags transactions that look unusual compared with training history.
Important note
Your notebook used one-hot encoded categories with drop_first=True. In this app, choosing Base / other means those encoded columns stay at zero, which matches the base category or any unseen category.
Cost Prediction
Efficiency Score
Anomaly Detection
Check for Anomaly
Use this after entering a completed transaction to see if it looks unusual.
Total amount (AED)
209.00
Transaction amount (AED)
208.00
Charged amount (AED)
219.00
Search
ENG
IN
12:03 PM
20-04-2026

Fig 23.

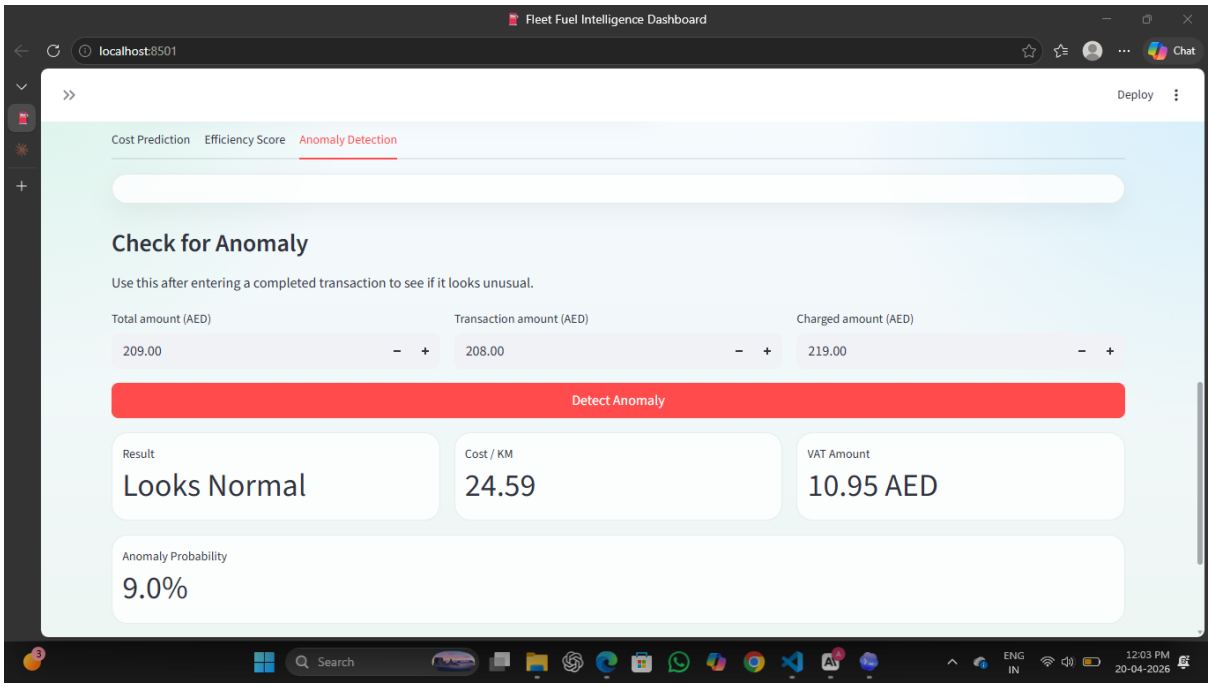

Fleet Fuel Intelligence Dashboard
localhost:8501
Chat
Deploy
Cost Prediction
Efficiency Score
Anomaly Detection
Check for Anomaly
Use this after entering a completed transaction to see if it looks unusual.
Total amount (AED)
209.00
Transaction amount (AED)
208.00
Charged amount (AED)
219.00
Detect Anomaly
Result
Looks Normal
Cost / KM
24.59
VAT Amount
10.95 AED
Anomaly Probability
9.0%
Search
ENG
IN
12:03 PM
20-04-2026

Fig 24.

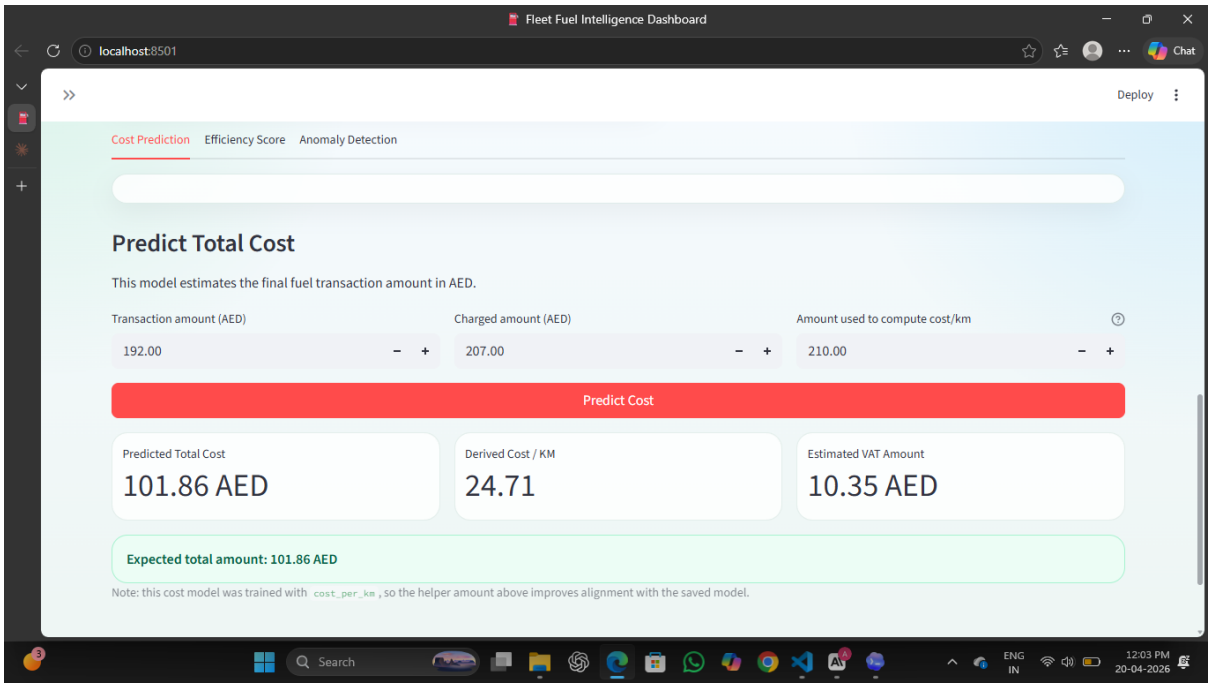

Fleet Fuel Intelligence Dashboard
localhost:8501
Chat
Deploy
Cost Prediction
Efficiency Score
Anomaly Detection
Predict Total Cost
This model estimates the final fuel transaction amount in AED.
Transaction amount (AED)
192.00
Charged amount (AED)
207.00
Amount used to compute cost/km
210.00
Predict Cost
Predicted Total Cost
101.86 AED
Derived Cost / KM
24.71
Estimated VAT Amount
10.35 AED
Expected total amount: 101.86 AED
Note: this cost model was trained with cost_per_km , so the helper amount above improves alignment with the saved model.
Search
ENG
IN
12:03 PM
20-04-2026

Fig 25.

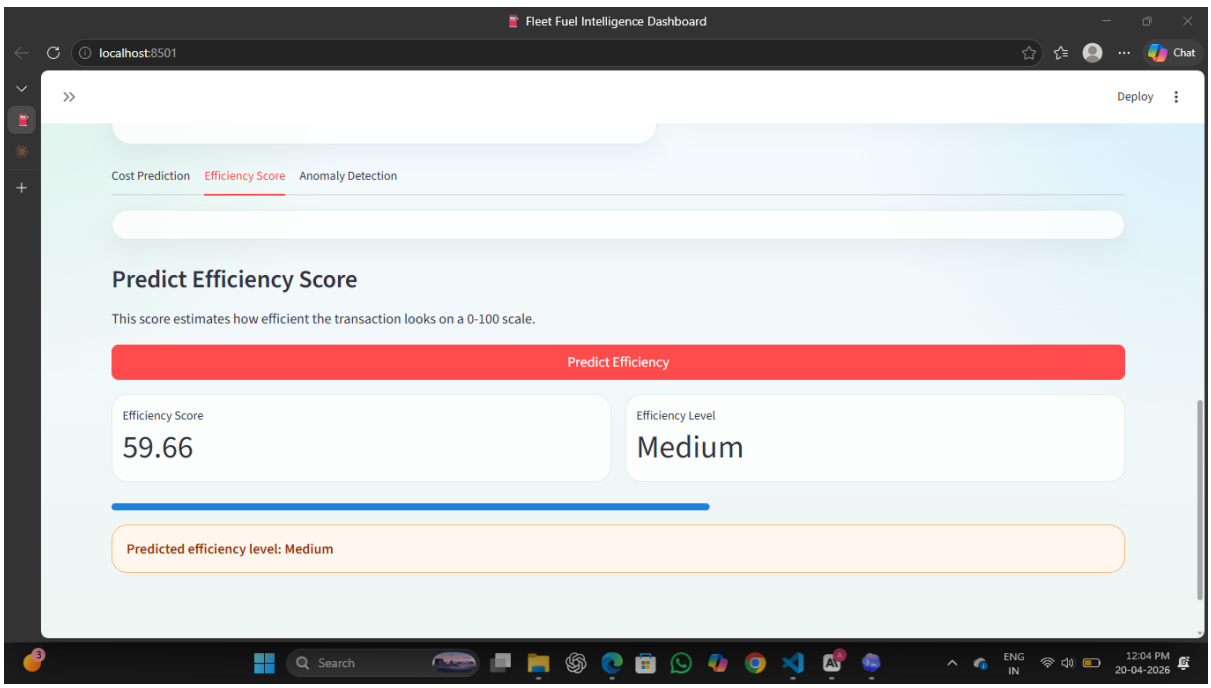

Fleet Fuel Intelligence Dashboard
localhost:8501
Chat
Deploy
Cost Prediction
Efficiency Score
Anomaly Detection
Predict Efficiency Score
This score estimates how efficient the transaction looks on a 0-100 scale.
Predict Efficiency
Efficiency Score
59.66
Efficiency Level
Medium
Predicted efficiency level: Medium
Search
ENG
IN
12:04 PM
20-04-2026

Fig 26.

CHAPTER 17: APPLICATIONS

17.1 Industrial applications

The proposed machine learning-based Fleet Fuel Intelligence system has significant applications across various industrial sectors where fuel management plays a critical role.

• Fleet Management Systems
Organizations with large vehicle fleets can use this system to monitor fuel consumption, predict costs, and evaluate efficiency. This helps in optimizing operations and reducing fuel wastage.

• Transportation and Logistics
Logistics companies can analyze fuel usage patterns and improve route planning, leading to reduced operational costs and improved efficiency.

• Fuel Monitoring Systems
Oil and fuel companies can integrate this system to monitor transactions and detect irregularities in fuel distribution.

• Fraud and Theft Detection
The anomaly detection model helps in identifying suspicious transactions such as fuel theft, leakage, or manipulation of records.

• Operational Efficiency Optimization
Industries can use efficiency scores to identify underperforming vehicles and take corrective actions.

17.2 Real –World Use cases

The system can be applied in real-world scenarios such as:

• Smart Fleet Management
Companies can monitor real-time fuel usage and improve decision-making.

• Public Transportation Systems
Government agencies can track fuel usage in buses and reduce unnecessary expenditure.

• Fuel Stations
Fuel stations can analyze customer transactions and optimize fuel supply.

• Energy Management Systems
The system can be integrated with energy platforms to improve sustainability and reduce fuel consumption.

• IoT-based Systems
The system can be connected with IoT sensors for real-time fuel monitoring and automated data collection.

• Industrial Fleet Operations at Nurol
The proposed system can assist in monitoring fuel consumption, detecting anomalies, and optimizing operational efficiency in large-scale industrial fleet environments.

CHAPTER 18: ADVANTAGES AND LIMITATIONS

18.1 Advantages

The proposed system offers several advantages:

• **Improved Accuracy**
Machine learning models provide accurate predictions of fuel cost and usage patterns.

• **Automation**
The system automates data analysis, reducing manual effort and errors.

• **Efficiency Evaluation**
The efficiency scoring model helps in identifying high and low-performing transactions.

• **Anomaly Detection**
The system effectively detects unusual transactions such as fuel theft or abnormal usage.

• **Scalability**
The pipeline-based architecture allows the system to handle large datasets and be extended for future use.

• **Real-Time Insights**
With dashboard integration, users can get instant results and insights.

• **Cost Optimization**
Helps in reducing fuel wastage and improving resource utilization.

18.2 Limitations

Despite its advantages, the system has certain limitations:

• **Dependence on Data Quality**
Poor or incomplete data can affect model performance.

• Model Complexity
Machine learning models require proper tuning and expertise.

• Computational Requirements
Training models on large datasets may require higher processing power.

• Limited Real-Time Processing
The current implementation may not fully support real-time streaming data.

• Overfitting Risk
Models may overfit if not properly validated.

• Maintenance Requirement
Models need periodic retraining with updated data for better performance.

CHAPTER 19: FUTURE SCOPE

19.1 Improvements

Although the current system provides effective results, there are several areas where improvements can be made to enhance its performance and usability.

- **Use of Advanced Machine Learning Models**
  More advanced models such as Deep Learning, LSTM, or Transformer-based models can be used to improve prediction accuracy, especially for time-series data.
- **Real-Time Data Processing**
  The system can be enhanced to process real-time data using streaming technologies, enabling instant analysis and faster decision-making.
- **Improved Feature Engineering**
  Additional features such as weather conditions, traffic data, and vehicle load can be included to improve model performance.
- **Model Optimization**
  Hyperparameter tuning and optimization techniques can be applied to improve the efficiency and accuracy of machine learning models.

19.2 Future Enhancements

The system can be further extended with additional features and functionalities:

- **Integration with IoT Devices**
  Sensors and IoT devices can be used to collect real-time fuel and vehicle data, making the system more dynamic and automated.
- **Development of a User Interface**
  A web or mobile application can be developed to visualize results,

making the system more user-friendly.

- **Cloud-Based Deployment**
  The system can be deployed on cloud platforms to handle large-scale data and provide remote access.
- **Automated Alert System**
  Alerts can be generated automatically when anomalies are detected, helping in quick response and prevention of losses.
- **Integration with GPS Systems**
  Combining fuel data with GPS tracking can provide deeper insights into vehicle movement and fuel usage patterns.

CHAPTER 20: CONCLUSION

The project “Fleet Fuel Intelligence: Cost Prediction, Efficiency Scoring and Anomaly Detection Using Machine Learning” successfully demonstrates the use of real-time industrial dataset support provided by **Nurol** enhanced the practical implementation and real-world applicability of the system.

The system integrates multiple components into a unified pipeline, including cost prediction, efficiency scoring based on mileage, and anomaly detection. This combined approach provides a more comprehensive and practical solution compared to traditional systems.

The cost prediction model accurately estimates fuel expenses, while the efficiency scoring model evaluates performance using mileage and consumption patterns. The anomaly detection model effectively identifies unusual transactions, helping in detecting inefficiencies and potential fraud.

The use of data preprocessing, feature engineering, and exploratory data analysis further improves the accuracy and reliability of the system. The integration of these components into an interactive dashboard enhances usability and provides real-time insights.

Overall, the project highlights the importance of data-driven decision-making in modern industries. It demonstrates that machine learning can significantly improve fuel management systems by increasing efficiency, reducing costs, and enhancing monitoring capabilities.

The proposed system is scalable, efficient, and suitable for real-world

applications. With future improvements such as real-time integration and advanced models, it can be developed into a fully automated intelligent fuel management solution.